King’s College London

Energy Electronics in the UK

UK Household Appliance Energy Consumption: A Bottom-Up Modelling Analysis

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Abstract

The United Kingdom's 2050 net-zero ambition is threatened by a critical "granularity gap" in its residential energy data, where official statistics lack the device-specific detail required for effective appliance regulation. This knowledge gap prevents policymakers from formulating evidence-backed strategies to manage domestic energy consumption, which leads this study to address a central question: Can a reproducible, UK-wide bottom-up model accurately estimate the annual operational electricity consumption and corresponding emissions for the 26 most common household devices? To answer this, this study employs a bottom-up engineering methodology, implemented via a Python-based model that systematically calculates national energy demand by aggregating the consumption of individual appliances using validated data on device stock, power ratings, and average usage patterns. The model's outputs, benchmarked against the official 'Energy Consumption in the UK' (ECUK) dataset, reveal that the 26 modelled appliances consume approximately 52.4 TWh annually, with a distinct hierarchy of impact dominated by cold appliances (Fridges/Freezers), high-power heating devices (Kettles), and entertainment devices (LCD Televisions). The analysis further demonstrates that for individual households, devices with long operational cycles (e.g., Dishwashers) or high-power draws (e.g., Gaming Consoles) are more energy-intensive than their national ranking might suggest. Ultimately, the findings confirm that such appliance-level insights are crucial for designing targeted and effective energy policies, directly answering the research question and fulfilling the project's core objectives.

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Acronyms

ABM	Agent-Based Model
BREDEM	Building Research Establishment Domestic Energy Model
CCC	Committee on Climate Change
DCMS	Department for Digital, Culture, Media & Sport
DESNZ	Department for Energy Security and Net Zero
DfT	Department for Transport
DSIT	Department for Science, Innovation and Technology
DUKES	Digest of UK Energy Statistics
ECUK	Energy Consumption in the
UKEDF	Électricité de France
GHG	Greenhouse Gas
GSMA	Global System for Mobile Communications Association
ICT	Information and Communications Technology
IEA	International Energy Agency
IoT	Internet-of-Things
MTP	Market Transformation Programme
ONS	Office for National Statistics
EDF	Électricité de France

List of Symbols

Notation

C_{bat}	Battery capacity	(Wh)
C_{device}	Annual operational emissions per device	(kg CO_2e/device/yr)
C_{nat}	Annual national operational emissions	(kt CO_2e/yr)
CF_{grid}	Grid carbon factor	(kg CO_2e/kWh)
$E_{act,device}$	Annual active-mode energy per device	(kWh/device/yr)
E_{device}	Total annual energy per device	(kWh/device/yr)
E_{nat}	Total annual national energy	(GWh/yr)
$E_{stby,device}$	Annual standby-mode energy per device	(kWh/device/yr)
E_{total}	Total energy consumption across all appliances	(kWh/yr)
h_i	Annual hours of use for appliance i	(h/yr)
N_i	Number of units for appliance i	(units)
N_{units}	Stock of a device installed in UK homes	(million)
n_{cycles}	Full charge cycles per day	(cycles/day)
P_i	Rated power for appliance i	(W)
P_{fleet}	Fleet-average on-mode power	(W)
P_{mid}	Mid-case on-mode power	(W)
P_{stby}	Standby power	(W)
T_{act}	Active minutes per day	(min/day)

1 Introduction

1.1 The Unseen Footprint of the Modern Home

In the race to achieve its ambitious 2050 net-zero target, the United Kingdom faces a formidable challenge that resides not in industrial smokestacks or distant power plants, but hidden in plain sight within its millions of homes. The modern household is a hub of technological activity, powered by an ever-growing array of electronic devices. While these appliances provide convenience and connectivity, they collectively create a significant energy footprint. In 2022, the UK residential sector accounted for 17% of total UK carbon dioxide emissions, totalling 56.4 million tonnes, and with the rising electronics adoption trends, this statistic will keep increasing if left unchecked [1].

The solution to curbing this rising demand lies in the hands of policymakers, who can implement targeted regulations on the most energy-intensive devices. However, effective policies cannot be based on theory alone; they require precise, granular data. Within the lifecycle of most electronics, there are two types of carbon emissions. The first type is embodied emissions; these are produced during the extraction of raw materials, manufacturing, transportation, and the installation of the device. The second type of emissions is operational emissions, which are generated when the electronic device is connected to the mains, either during use or while charging [2]. This study focuses on the latter, as understanding the operational energy consumption of household appliances is the first and most critical step toward meaningful regulation.

1.2 The Granularity Gap in UK Energy Policy

While the UK government has demonstrated a clear commitment to its net-zero goals, a critical knowledge gap persists in the official evidence base. High-level reports from bodies like the Committee on Climate Change (CCC) outline broad sectoral targets, but when it comes to the specific impact of individual household appliances, from washing machines to microwaves, the data regarding emissions and energy is limited [3]. Successful policies depend on context, not just theory. Official datasets such as the Energy Consumption in the UK (ECUK) or the Digest of UK Energy Statistics (DUKES) fail to provide a comprehensive list of ownership-adjusted energy levels, including device models and their corresponding units in the residential sector. Another evident issue is that these datasets do not include the variations in certain appliances. A survey done in 2022 reports that 48% of adults have cut their oven usage by half, after switching to air fryers [4].

This "granularity gap" has profound consequences. Without a detailed, bottom-up understanding of which specific appliances are the worst offenders, policymakers risk designing regulations that are inefficient, misdirected, or fail to address the true drivers of domestic energy consumption. To effectively steer the UK towards its climate goals, three linked challenges must be tackled before policymakers can target the worst offenders. First, the exact stock of each electronic device in UK households must be established. Second, both active-mode energy consumption and standby power draw need to be measured, because many appliances consume electricity around the clock even when not in use. Third, reliable data on how long devices operate in each mode, viewing, gaming, charging, sleep, or standby, must be gathered. Once accurate stock numbers, mode-specific wattages and time-of-use patterns are in place, operational emissions can be calculated robustly, giving policymakers the evidence required to set minimum-efficiency standards or stand-by caps that steer the UK toward its climate goals.

1.3 A Bottom-Up Solution

This project aims to fill this critical knowledge gap by developing and implementing a transparent, bottom-up energy simulation model. Using Python, this model quantifies the operational electricity consumption and carbon emissions of 26 of the most common household appliances by integrating national ownership data, device-specific power ratings, and average usage times. This approach enables evidence-based regulations to be implemented for high-priority devices, guided by systematic emission rankings. The model will also be benchmarked against the ECUK 2023 data sets to validate its outputs and to compare the project's bottom-up methodology to ECUK's hybrid framework. Key innovations of this study include an open-source framework that is adaptable to other regions and includes ownership-weighted consumption metrics. The methodology is also scalable for future expansion to include new electronics and device models.

1.4 Report Structure

The remainder of this report is organised as follows. Chapter 2 presents a critical review of residential energy literature, focusing especially on the ECUK 2023 methodology, to identify where existing studies fall short at the appliance level. Chapter 3 formalises the research problem, lays out the guiding research questions and states the three objectives. Chapter 4 then describes in detail the bottom-up, Python-based modelling methodology: how device stocks, runtimes, and power draws feed into a Monte-Carlo engine to produce annual energy and CO_2e outputs. Chapter 5 presents those results, ranking the 26 devices by energy use and operational emissions, benchmarking them against the ECUK data to test model validity, and highlighting the high-priority appliances. Finally, Chapter 6 draws conclusions, discusses policy implications and suggests avenues for future work.

2 Literature Review

2.1 Introduction

This chapter reviews the literature on household plug-load appliances, specifically, cold appliances, wet appliances, audio-visual/IT equipment, cooking equipment, lighting, and miscellaneous small plug-in devices. Larger thermal systems such as space or water heating are excluded, because they depend on different physical drivers, data sources and policy instruments.

Residential plug-load electricity-demand modelling is singled out in recent UK decarbonisation analyses as “central to developing an energy strategy to support delivery of net-zero” [5]. At the appliance level, global evidence shows that well-designed efficiency policies can halve the electricity consumption of major household devices without raising purchase prices [6]. Capturing such device-level impacts is therefore crucial when the UK sets minimum-efficiency standards, designs market-transformation programmes, and prepares demand-side-flexibility rules for smart appliances.

The review begins by comparing the established paradigms of top-down, bottom-up, and hybrid modelling approaches to justify the methodology selected for this study. It then quantifies the scale and characteristics of UK household electricity use to establish the policy relevance of the research problem. Subsequently, it provides a critical audit of the official data sources, focusing on the ECUK publication, to identify the specific limitations that create a knowledge gap. Finally, the chapter identifies these residual gaps and positions the present study as a direct response, before summarising the key insights that inform the methodology detailed in Chapter 3.

2.2 Modelling Paradigms in Residential Energy Demand

2.2.1 Top-down macro models

Top-down tools embed household electricity demand in an economy-wide computable-general-equilibrium or econometric framework. Demand is described by an aggregate function of income and energy prices that moves in step with the rest of the macro-economy [7]. Such models capture feedbacks, for example, a rise in appliance prices can depress spending elsewhere, and stay fully consistent with the GDP, trade balances and fiscal flows.

For device policy, two weaknesses dominate. First, top-down models keep only very coarse technology detail; a single sector (household electrical goods) covers everything from televisions to refrigerators, so individual appliances cannot be isolated [8]. Second, physical parameters such as rated power or duty cycle are exogenous. Tightening the efficiency standard for fridges must therefore be imposed from outside rather than calculated inside the model.A pure top-down approach is thus unsuitable for ranking individual plug-load appliances or for exploring uncertainty around their parameters.

2.2.2 Bottom-up engineering models

Bottom-up models work in the opposite direction, summing demand across every appliance stock unit:

$$E_{\text{total}} = \sum_{i} P_i \, h_i \, N_i,$$

Where P_i is rated power, h_i annual hours of use and N_i the number of units for appliance i [9]. This explicit treatment makes them the natural choice for minimum-efficiency standards, standby caps and time-of-use incentives, because each parameter can be varied independently. Their key drawback is data burden – thousands of stock vintages, run-time profiles and in-use wattages must be collected, and they seldom embed macro feedbacks such as income effects. Later chapters, therefore, add a price-elasticity sensitivity sweep to bridge that gap.

2.2.3 Hybrid and Agent-Based Approaches

To combine the strengths of both paradigms, researchers have developed hybrid models. A common approach is to "soft-link" a bottom-up sectoral model with a top-down economy-wide model; for example, the UK's Centre for Research into Energy Demand Solutions (CREDS) uses detailed bottom-up analyses of energy service demands as inputs to the UK TIMES model to construct coherent net-zero scenarios [10].

Agent-based models (ABMs) offer another route to enhancing bottom-up approaches by simulating the emergent, system-level consequences of individual, heterogeneous behaviours. Rather than assuming a representative consumer, an ABM can model a diverse population of households whose interactions and decisions generate realistic energy demand profiles [11]. A recent behavioural ABM, validated against real-world transformer data, was able to reproduce neighbourhood load profiles with a total energy-use variation typically within 1-4% [12].

This dissertation adopts a pure bottom-up Monte-Carlo model for transparency and data tractability, but it borrows lessons from the wider literature by incorporating a sensitivity analysis to identify key drivers and a Monte-Carlo simulation to propagate parameter uncertainty, addressing the limitations of a purely deterministic approach.

2.3 The UK Residential Sector: A Key Area for Decarbonisation

To contextualise this study, it is essential to quantify the scale of household energy use in the UK. In 2023, the residential sector accounted for 26% of the nation’s final energy consumption. While total household energy use has trended downward over the past decade, driven by better building insulation and more efficient appliances, the sector remains a primary focus of national climate policy [13].

When considering electricity specifically, the energy vector central to this dissertation, the role of plug-load appliances becomes particularly pronounced. In 2023, UK households consumed approximately 93 terawatt-hours (TWh) of electricity across 28.6 million homes [14]. While space and water heating dominate total household energy use (including natural gas), a breakdown of a typical electricity bill reveals a different picture. According to analysis by the Energy Saving Trust, plug-load appliances collectively account for the majority of electricity consumption: wet appliances (washing machines, etc.) are responsible for 14% of a typical electricity bill, cold appliances for 13%, consumer electronics for 6%, lighting for 5%, and cooking appliances for 4% [15] [16].

2.4 Data and Evidence Base for UK Appliance Demand

2.4.1 Energy Consumption in the United Kingdom (ECUK)

2.4.1.1 Overview of the ECUK

Energy Consumption in the UK is the Department for Energy Security and Net Zero (DESNZ) annual, accredited statistical release. It has published appliance-level electricity estimates continuously since the 1970s, making it the longest time-series of its kind in the UK [17].

2.4.1.2 Key Household Data in ECUK

For households, ECUK provides (i) sector totals in the main PDF summary and (ii) three Electrical Products tables, A1 (Electricity consumption by domestic appliances 1970 to 2023), A2 (Number of appliances owned by households in the UK 1970 to 2023) and A3 (Average consumption by household electrical appliances 1970 to 2023). [18].

2.4.1.3 Modelling Approach (Hybrid Top-Down/Bottom-Up)

ECUK is a hybrid system: sector totals are derived from a top-down energy balance, while the appliance tables are generated from a bottom-up stock model [19].

2.4.1.4 Data Sources and Workflow

The stock model is built from three inputs: historic ONS sales aged with DEFRA Market Transformation Programme lifetime curves (to give the number of units in use) [20]; in-use power levels measured in MTP laboratory tests [21]; and hourly usage profiles taken from BREDEM-2012, itself calibrated to the 2014 UK Time-Use Survey [22]. Annual kWh for each appliance category is calculated as stock × watts × hours and published in the workbook. The header states these tables are “independently modelled and not compatible with DUKES electricity consumption,” confirming they are not reconciled to the top-down total [23].

2.4.1.5 Limitations for Device-Level Analysis

First, the product groupings are coarse: devices such as smart speakers and Wi-Fi routers are bundled into “CONSUMER ELECTRONICS”, which hides fast-growing end-uses [23]. Second, ECUK warns that some behavioural research dates to 2007, so modern trends, streaming, and home working, “may not be fully captured” [18]. Finally, ECUK publishes point estimates only; it provides no uncertainty band around its appliance totals.

2.4.1.6 Implications for This Dissertation

ECUK supplies a national benchmark. Where the project’s Monte-Carlo model aligns with ECUK, it lends credibility to the granular approach; where it diverges, especially for fast-moving categories such as AV/IT, the comparison identifies data gaps that future surveys or smart-plug trials should target.

2.4.2 Environmental Impacts of the UK Digital Sector Report

The most comprehensive recent attempt to quantify appliance-related electricity use at a national scale is the “Environmental Impacts of the UK Digital Sector - Data Driven Report” (May 2024) [24]. Commissioned by DCMS and DSIT, the study compiles a bottom-up inventory of user devices, data centre equipment, and network hardware, then combines these stocks with in-use wattages and embodied emissions factors to build a national footprint.

Device stocks are reconstructed from historic sales and market-share data; power draws come from laboratory test datasets, while usage hours are benchmarked against UK time-use surveys. Electricity demand is calculated device-by-device and summed into three domains, user devices, data centres and networks, mirroring the approach adopted in this dissertation for household plug-loads. The report estimates a sector-wide annual electricity demand of 31.35 TWh, roughly 9.8 % of total UK demand, with user devices responsible for one-quarter of that total.

- User devices, particularly smartphones, laptops, and TVs, dominate the digital sector’s embodied emissions (~50% of total GHGs), though electricity use is more distributed (e.g., computers and TVs consume the most power).”

- The authors highlight the lack of UK-specific runtime data and rapid device turnover as major uncertainties, recommending regular surveys and compulsory reporting to keep inventories current.

    - **Policy salience:** Plug-load appliances, even within a broader digital context, constitute a double-digit share of national electricity; fine-grained modelling is therefore policy-relevant.

    - **Data refresh needed:** The call for UK-specific runtime datasets aligns with this dissertation’s use of stochastic Monte-Carlo sampling to propagate uncertainty in hours-of-use.

Where the study aggregates all user devices, the present work drills down to individual household categories (cold, wet, AV/IT, cooking, lighting), providing a complementary level of detail.

2.4.3 Policy Driver Example – EU / UK Standby-Power Regulations

Standby-power limits introduced under the EU Ecodesign framework, and retained in UK law after Brexit, show how a single device-specific regulation can cut national electricity demand [25]. Commission Regulation (EC) 1275/2008 was implemented in two stages: from 2010, it capped standby power at 1.0 W (2.0 W with a status display), and from 2013, these limits were tightened to 0.5 W and 1.0 W, respectively [26]. This was amended in 2013 by Regulation (EU) 801/2013, which imposed caps on “networked-standby” that, depending on the product, fell to between 3 to 8 W from 2017 and 2 to 8 W from 2019 [27]; as a result, future appliance policies must target active-mode efficiency and usage patterns, the very parameters the project’s bottom-up model explores.

2.5 Gap Analysis & Research Positioning

This dissertation delivers a confidence-banded, device-level estimate of 2025 UK plug-load electricity demand. The present model still relies on fixed average hours of use. For example, a primary television is assumed to run about four hours per day, and it does not yet incorporate macro feedbacks such as price elasticity. This leads to potential rebound effects remaining outside the analytical boundary. Closing these gaps forms a clear future agenda: first, gathering fresh UK run-time data via smart-plug trials or an updated Time-Use Survey, and second, linking the appliance model to an elasticity module that captures income- and price-driven responses. Even with these limitations, the study advances the evidence base by rebuilding stocks for twenty-six appliances, propagating technical uncertainty through 10,000 Monte-Carlo iterations, and using a ±10 % sensitivity analysis to identify which input parameters exert the greatest influence on national plug-load demand, information that can guide policymakers toward the most impactful data-collection priorities.

3 Problem Formulation

3.1 Problem Definition

The UK's residential sector alone consumed 93 terawatt-hours of electricity, which is 35% of the total national consumption [28]. Although publicly available statistics are available, they are at the end-use level (e.g., cooking, refrigeration), lacking specific model-level data, and don't include all the appliances that most households use. This granularity gap limits evidence-backed regulations and negatively impacts eco-design standards on high-priority devices.

Despite the updates to the ECUK 2023, the datasets do not include most appliances, such as air fryers and toasters, among others. Additionally, they are broad and lack detail on specific models that society uses, instead relying on a methodology that still uses data from 2007.

Research Question: Can a reproducible, UK-wide model estimate annual operational electricity consumption and its corresponding operational emissions for the 26 most common household devices?

3.2 Research Questions

- **What** is the annual estimate of the 26 most common devices regarding their energy consumption and operational emissions?
- **How** can a deterministic model derive those values from stock, power-rating, and usage data?
- **When** key parameters vary within realistic boundaries, how sensitive are the rankings?
- **Why** do appliance-level insights matter for evidence-backed regulations and sustainable strategies to aid the UK's climate goals

3.3 Objectives

1. Compile a validated device stock database for 26 household devices
2. Implement a Python-based model that outputs annual energy consumption, operational emissions
3. Rank appliances by emissions and identify at least two high-priority devices, where a 20% decrease in energy consumption would save >0.2 Mt CO_2e/yr

3.4 Assumptions, Limitations & Scope

Usage patterns: The daily usage limits represent the annual averages; the seasonal variations that affect usage patterns are not encapsulated. For the usage patterns, the active use of the devices is considered as well as their standby energy consumption. However, for electronics that are battery-powered, their energy consumption will be modelled using their battery capacity.

Device Homogeneity: Each device is represented by a single 'typical' model (Monte Carlo simulation for model variations)

Grid Emissions: A constant grid factor of 0.225 kg CO_2e kW/h was used; Diurnal variation was ignored.

Boundary: 26 household devices only; embodied emissions excluded

Validation data: The ECUK does not include all appliances, thus a risk of inaccuracies

3.5 Success Criteria

Model success is judged against three practical checks:

1. **Internal coherence:** Appliance stocks, runtimes, and wattages reconcile to a 2025 national total that sits within the long-term trend band reported in ECUK's Electrical-Products tables.

2. **External plausibility:** Where ECUK provides robust device-level figures, the model matches them to within a small tolerance; larger gaps are accepted only when they can be traced to clearly documented differences in behavioural or technical assumptions.

3. **Uncertainty transparency:** Every active-power input carries an explicit ±10 % triangular range that is propagated through a 10,000-draw Monte-Carlo analysis, and the resulting P5–P95 bounds are reported alongside each headline result.

4 Methodology

4.1 Overview of Modelling Approach

This study employs a bottom-up energy modelling framework to quantify the electricity consumption and corresponding operational carbon emissions of household appliances in the UK. This approach is executed through four sequential phases, as illustrated in Figure 4.1.

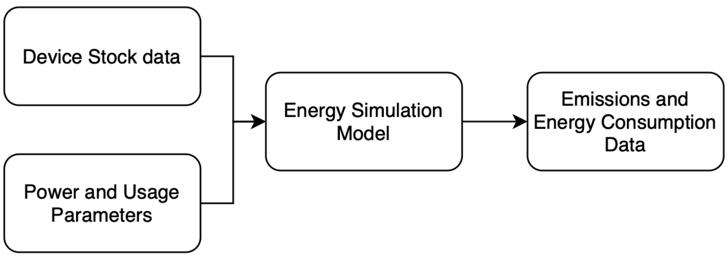

FIGURE 4.1: OVERVIEW OF THE FOUR-PHASE BOTTOM-UP MODELLING FRAMEWORK

First, a Device Stock Assessment is conducted to create an index of 26 common household appliances and compile their corresponding national ownership data. Second, a Power and Usage Parameter analysis determines the active and standby power drawn by each appliance, along with their average daily usage patterns. Third, an Energy Simulation model calculates the annual energy consumption at both the individual device and national levels. Finally, an Emissions Assessment converts the calculated energy consumption to operational carbon emissions (CO_2e) using the UK's national grid carbon factor.

4.2 Core Energy Model Formulation

The bottom-up calculator transforms four primary inputs for every appliance, as defined in Table 4.1.

Symbol	Definition	Units
P_{mid}	mid-case on-mode power	W
P_{stby}	stand-by power	W
T_{act}	active minutes per day	min / day
N_{units}	stock installed in UK homes	million

4.1 DEFINITION OF PRIMARY INPUT PARAMETERS FOR THE CORE ENERGY MODEL

The grid-carbon factor is fixed at: $CF_{\text{grid}} = 0.22535 \text{ kg CO}_2\text{e / kWh}$

The following equations form the core of the model:

1. **Annual active-mode energy per device:** Converts on-mode power from Watts to kW, applies the daily run-time in hours, and scales to one year.

- $E_{\text{act,device}} = \left(\frac{P_{\text{mid}}}{1000}\right) \times \left(\frac{T_{\text{act}}}{60}\right) \times 365$ [kWh / device / yr]

2. Annual standby-mode energy per device: Calculates the energy consumed during the minutes per day when the appliance is idle.

- $E_{\text{stby,device}} = \left(\frac{P_{\text{stby}}}{1000}\right) \times \left(\frac{1440 - T_{\text{act}}}{60}\right) \times 365$ [kWh / device / yr]

3. Total annual energy per device: Sums the active and standby mode-specific loads for a single device.

- $E_{\text{device}} = E_{\text{act,device}} + E_{\text{stby,device}}$ [kWh / device / yr]

4. Annual operational emissions per device: Converts the annual energy consumption of one device (kWh) to kilograms of CO_2e

- $C_{\text{device}} = E_{\text{device}} \times \text{CFgrid}$ [kg CO2 / device / yr]

5. National energy: Scales the total energy per device by the national stock, converting from kWh to GWh.

- $E_{\text{nat}} = \frac{E_{\text{device}} \times N_{\text{units}} \times 1{,}000{,}000}{1{,}000{,}000}$ [GWh / yr]

6. National operational emissions: Applies the same carbon factor to the national energy figure, converting from kg to kilotonnes.

- $C_{\text{nat}} = \frac{E_{\text{nat}} \times CF_{\text{grid}} \times 1\,000\,000}{1\,000}$ [kt CO2 / yr]

7. ±10% active-power bounds (error bars): Provides the lower and upper whiskers for the uncertainty analysis plotted in the final charts.

- $E^{\min}_{\text{act,device}} = 0.9 \times \frac{P_{\text{mid}}}{1000} \times \frac{T_{\text{act}}}{60} \times 365$ [kWh / device / yr]

- $E^{\max}_{\text{act,device}} = 1.1 \times \frac{P_{\text{mid}}}{1000} \times \frac{T_{\text{act}}}{60} \times 365$ [kWh / device / yr]

4.3 Validation and Uncertainty Analysis

To ensure the robustness and credibility of the model, several measures for validation and uncertainty analysis were implemented.

4.3.1 Monte Carlo Uncertainty for Power Rating

A Monte Carlo uncertainty propagation was conducted to assess how uncertainties in the input parameters affect the model's output [29]. For each device, active-mode power was sampled 10,000 times from a triangular distribution bounded at ±10% of the manufacturer-reported most likely active power values. Triangular sampling was chosen because manufacturers typically quote a single nominal wattage together with plausible minima and maxima; the distribution emphasises that most likely point while still accounting for extremes, without the heavy tails of a normal distribution [30]. Daily usage times, standby power, and ownership rates were held constant, so the experiment isolates the impact of active-power uncertainty on national and device-level electricity demand.

4.3.2 Validation Against ECUK Benchmark

Device-level electricity totals are benchmarked against the Electrical-Products tables in Energy Consumption in the United Kingdom (ECUK). While this provides a nationally recognised reference, the ECUK figures are themselves modelled rather than metered: they combine BREDEM-2012 usage profiles, appliance-ownership surveys and laboratory power tests, then upscale to the national stock. ECUK's estimates use a hybrid framework, so a comparison with this study's bottom-up model can reveal methodological differences, not just parametric ones. This provides further justification for using ECUK as the benchmark.

4.3.3 Sensitivity Analysis and Reproducibility

A sensitivity analysis was performed to evaluate the influence of input parameter variations on the model's output. Each input parameter (power, active daily usage, ownership) was independently varied by ±10% while the others were held constant; the resulting outputs are presented graphically in Chapter 5. The model also ensures reproducibility by explicitly defining all inputs, allowing for adjustments such as the use of different regional carbon factors. All calculations are deterministic; therefore, there are no random elements outside of the Monte Carlo simulation.

4.4 Device-Specific Modelling Assumptions and Nuances

To construct a deterministic model from complex real-world behaviours, a number of simplifying assumptions and engineering judgments were required. This section details the most significant device-specific nuances that were standardised for the model.

- **Deterministic Usage Patterns:** A core assumption of the model is the use of fixed, deterministic usage times for all devices. These times are held constant across the Monte Carlo runs because robust, UK-wide statistical distributions for appliance runtimes are not yet available. This simplification is a primary reason for flagging primary data collection on runtimes as a future research priority.

- **Standby Power Assumptions:** The model assumes that standby power for relevant devices adheres to the current, tight Ecodesign limits. However, it is a known nuance that older, pre-regulation devices still in circulation may have standby power draws that exceed these caps, which could lead to the model underestimating the true national total for standby consumption.

Kitchen and Cooking Appliances:

- **Kettles:** The model assumes a fixed daily runtime based on average use. In reality, the energy consumed per use is highly dependent on the volume of water being boiled, a variable not captured in this model.

- **Washing Machines and Dishwashers:** A single average power rating is used for these devices. This does not account for the significant variation in energy consumption caused by different cycle settings (e.g., eco-mode vs. intensive wash), load sizes, or water temperatures (e.g., a 60°C cotton cycle consumes more than a 30°C cycle).

- **Fridge-Freezers:** The model assumes a constant compressor duty cycle. Real-world energy consumption is influenced by factors such as the ambient temperature of the room, how frequently the door is opened, and the thermal mass of the food inside, none of which are explicitly modelled.

Entertainment and media

- **Televisions:** The model does not account for variations in on-mode power caused by different screen brightness settings selected by the user.

- **Gaming Consoles:** The model uses a weighted average power based on the market share of different consoles. This does not capture the dynamic power draw of a single console, which varies significantly depending on the computational intensity of the specific game being played.

Personal (battery-powered) electronics

- **Charging Cycles:** A core assumption for all battery-powered devices is that they undergo exactly one full charge cycle per day. This simplifies the complex reality of user charging habits, where devices may be charged partially, overnight, or not at all on a given day.

- **Charging Efficiency:** The model assumes zero charging losses, meaning all energy drawn by the charger is transferred to the battery. In practice, some energy is always lost as heat during the charging process.

4.5 Device Input Parameter Selection

This section details the methodology used to determine the input parameters for each household device. To reflect the different calculation methods required, devices are organised into two primary categories.

4.5.1 Mains-Powered Devices

<u>Kitchen and Cooking Appliances</u>

Fridge-Freezer

The combined fridge-freezer is a ubiquitous appliance, with an estimated stock of 21.03 million units in the UK [23]. Its power consumption is characterised by a duty cycle rather than a simple on/off state. The compressor's active power draw is approximately 150W during the cooling cycle, while the idle standby power is 15W between the cooling cycles [31]. It is estimated that the compressor runs for a cumulative total of 480 minutes (8 hours) per day to maintain its target temperature, which defines its active usage time [32].

Kettles
An estimated 27 million kettles are present in UK households [23]. These devices operate in a binary mode; they draw their full rated power of approximately 3000W while boiling and consume no power when off, resulting in a standby power of 0W [31]. The typical daily usage time is estimated to be 12 minutes [33].

Dishwashers
Dishwashers are found in 14.2 million households [34]. Their active power during a cycle is approximately 800W, with a standby power of 0.9W [35][36]. Based on an Energy Saving Trust (EST) report indicating usage of four 90-minute cycles per week, the average daily active time was calculated to be 51 minutes [37][38].

Air Fryers
An estimated 16.5 million air fryers are used in UK households [39]. Their active power is rated at 1500W, with a standby power of 0.5W, compliant with Ecodesign regulations [40][41]. The daily usage time was estimated to be 25 minutes, based on a typical meal preparation cycle [42][43].

Electric Hobs
The stock of electric hobs is approximately 14.8 million units [23]. A single heating ring is rated at an active power of 1800W. The standby power is 1W, attributed to the small continuous draw from digital displays and controls on modern ceramic hobs [44][45]. The daily usage time is estimated at 20 minutes [38].

Microwaves
Microwaves are estimated at 25.6 million units in UK households [23]. They consume approximately 1000W during active use and have a standby power of 2W [38][46]. The daily usage time is estimated to be 11 minutes [47].

Electric Ovens
With a stock of approximately 20.9 million units, electric ovens operate on a duty cycle of around 25% [23]. While their peak power is 2200W, this duty cycle results in an average active power of 550W and the standby power is 2W [48][33]. Based on common cooking tasks, the daily active usage time was estimated to be 35 minutes [49].

Coffee Machines
Coffee machines are found in 16.2 million households [34]. The active power was determined to be approximately 1400W, calculated by averaging the power for the heating element (1360W) and the dispensing mechanism (1422W) [50]. The standby power was calculated to be 0.77W, based on an estimated annual standby cost of £1.76 at a rate of 26p/kWh [51]. The active usage time for heating and dispensing is 3 minutes.

Rice Cookers
There are an estimated 4.5 million rice cookers in the UK [39]. Based on measurements from 14 different devices, the average active power consumption is 700W, with a standby power of 0W [52]. The daily usage time is estimated to be 30 minutes, based on typical cooking durations [53].

Toasters

Toasters are present in 21.9 million households [39]. A typical two-slice model uses 900W during operation and has a standby power of 0W [16][45]. According to data from Électricité de France (EDF), the average daily usage is approximately 9 minutes [54].

Washing Machines

With a 97% household ownership rate, the total stock of washing machines in the UK is approximately 27.5 million [55]. While the peak power during the water heating phase can reach 2000W, an average active power of 700W was used for the model. This figure was derived from an analysis of energy consumption data across a range of typical models, from highly efficient A-rated machines to less efficient D-rated ones, to accurately represent the broad spectrum of devices in use [38]. Standby power is rated at 1W. Based on a one-hour cycle used four times per week, the average daily active usage time is 34 minutes [37][38].

Computing and Networking Devices

Wi-Fi Routers

The ownership of Wi-Fi routers in UK households was estimated using data from Office of Communications (Ofcom), which states that 5% of the UK population does not have home internet access [56]. Applying this to the 28.4 million households reported by the Office for National Statistics (ONS) [57], and assuming one router per connected household, the total number of active routers is estimated to be 26.98 million. The power consumption was determined by averaging the idle (8.99W) and load (12.76W) power ratings of 14 common router models, resulting in an average usage value of 10.88W [58]. As industry guides suggest that routers are inexpensive to run and are typically left on continuously, a daily usage time of 24 hours was assumed [59].

Desktops

According to data from the Energy Consumption in the UK (ECUK): Electrical Products Tables, there are an estimated 3.4 million desktop computers in UK households [23]. The typical power consumption for these devices ranges from 50W to 180W; therefore, an average operational load of 100W was assumed for this study [60]. For standby and off-mode power, the value was set at 0.5W, in line with the European Union (EU) regulation 2023/826, which became effective on May 9, 2025 [61]. An analysis by EDF indicates that the average daily usage for a desktop computer is 138 minutes [54].

Monitors

The household stock of monitors is 19.2 million units, based on a report from Cambridge University [24]. To determine a representative power rating, the most common UK monitor was identified. Market analysis indicates Dell is a leading brand in the EU, and a review of major UK retailers' sales data reveals the 'Dell Monitor - S2725DS' as a consistently popular model [62]. According to its official specifications, this model has an on-mode power consumption of 21.4W and a standby power of 0.3W [63]. To ensure methodological consistency, the monitor's daily usage time was aligned with that of a desktop computer (138 minutes), as these devices are operated in conjunction.

Projectors

The number of projectors in UK households was extrapolated from an Information and Communications Technology (ICT) Impact Study by the European Commission, which reported 4 million units in the EU post-Brexit [64]. The UK's share was calculated based on its population relative to the EU27 (approximately 15%), resulting in an estimated stock of 0.6 million units in the UK [65]. The power rating was based on a representative model from Epson, the largest projector manufacturer in the UK [66]. Their most common home model, the 'EH-TW740', consumes 225W in Eco mode and 0.3W in standby [67]. The same European Commission report specified an average daily usage of 30 minutes for home theatre projectors, which was adopted for this study [64].

Printers
The number of printers in UK households was taken as 8.11 million [68]. With a market share of 34.2% in Q4 2024, Hewlett-Packard (HP) is the leading brand [69]. Their most common household model in the UK, the 'HP DeskJet 2810e', was selected as the representative device. This printer has an operational power draw of 26.64W and a standby power of 1.4W [70]. The daily usage time was calculated based on a benchmark test of printing 30 pages per month [71]. Using the printer's official printing speed, this corresponds to a daily active usage time of approximately 0.15 minutes.

<u>Entertainment and Media</u>
Home Consoles
Based on an analysis of Ofcom ownership data and ONS household figures, the estimated number of active home console units (PlayStation 5 and Xbox Series X/S) in UK households is 6.54 million [72]. This total comprises an estimated 3.88 million Xbox Series X/S units and 2.66 million PlayStation 5 units. Based on the proportional ownership of the 6.54 million PlayStation 5 and Xbox Series X/S consoles in the UK, the weighted average power consumption for a typical console is approximately 150 W during active gaming and 0.42 W when it goes to sleep mode after 10 minutes of no activity [73][74]. This average accounts for the different market shares and power needs of each console, including the powerful Xbox Series X and the more numerous, energy-efficient Xbox Series S, to provide a single representative figure for a UK games console. According to the EDF analysis, the home console's average household usage per day is 150 mins [54].

LCD TVs
The modelling of energy consumption for LCD televisions was based on the UK's stock of 52.3 million units, which was disaggregated into 26.3 million primary and 26.1 million secondary televisions [23]. The on-mode power for each sub-stock was determined by correlating market-share data on screen-size percentages with their corresponding manufacturer power ratings[103]. This methodology resulted in intrinsic on-mode power values of 63.7 W for primary sets and 36.2 W for secondary sets; a standby power of 0.5 W was applied universally [104][105][106]. By weighting these power values by the number of units in each sub-stock, a population-weighted operational power of 50.4 W was established for the overall fleet. The full calculation for this weighted power is provided in Appendix A. A separate parameter for usage, used to calculate total energy consumption, was defined by a daily viewing time of 270 minutes, based on data from Ofcom [102].

OLED TVs
OLED televisions were assumed to function as the primary television in a household, with a total stock of 1.05 million units [23]. To select a representative model, market data from Omdia (a technology research firm) was used, which identifies LG as the market leader for OLED sales from 2013-2023 [76]. Within LG's portfolio, a top-tier UK publication cites the 'LG C4' series as the most popular choice [75]. Consistent with the 55-inch assumption for primary TVs, this model's typical power consumption is 81W in on-mode and 0.5W in standby [75]. The daily usage time was taken as 270 minutes, in line with Ofcom's analysis for daily view time for televisions [102].

Set-Up Boxes

There are an estimated 26.04 million set-top boxes in UK households [23]. Given its significant market presence and popularity in the UK, the 'Sky Q' box was selected as the representative device for this category. This device has a typical on-mode power consumption of 20.1W and a standby power of 0.4W [77]. Its daily usage time is estimated to be 196 minutes [78].

Smart Speakers
The estimated stock of smart speakers in UK households is 9.37 million, a figure derived from a gov.uk report stating 33% household ownership [79] applied to the 28.4 million total UK households reported by the ONS [57]. To model energy consumption, the Amazon Echo was selected as the representative device, identified by an Ofcom Media Nations report as the UK's most used smart speaker [80]. Based on its specifications, this study considers two power states: an active mode of 2.4W and an idle mode of 1.3W [81]. The daily active usage time was calculated to be 36 minutes; this was derived by taking the average daily radio listening time of 176 minutes (from Radio Joint Audience Research (RAJAR), Q4 2023) [82], applying the 20% share of in-home audio listening attributed to smart speakers (from Ofcom) [80], and adjusting the resulting 35-minute figure slightly upwards to account for non-radio activities like music streaming. Therefore, the model assumes each device operates in active mode for 36 minutes and idle mode for the remaining 23 hours and 24 minutes daily.

4.5.2 Battery-Operated Devices

Devices in this category are grouped separately as their input parameters were determined using a distinct methodology compared to mains-powered equipment. While the total number of units for each device was calculated using standard stock modelling, their power consumption and usage times were handled differently. Power consumption was defined by two charger states: the power drawn from the mains while actively charging the device, and the idle power drawn by the charger when no load is present. The active usage time for each device was not taken from user reports but was instead calculated based on the device's battery capacity and its charger's power output, according to the methodology detailed in Appendix B and the assumptions outlined in Section 4.4.

Smart Phones
Based on 2025 estimates indicating that 95% of the UK population owns a smartphone, and assuming one phone per person, the total stock is 64.93 million units [83]. The representative model was identified as the iPhone 13. This selection was based on Apple's market leadership, which held the top spot globally in Q1 2025 and a 48.05% share of the UK's mobile vendor market [84][86]. Furthermore, the iPhone 13 itself accounted for 16.47% of the market in 2025, making it the most-used model [85]. The parameters were therefore based on this phone's battery capacity of 12.41Wh [87]. A standard, non-fast charging unit was selected as the representative charger, with a charging power of 5W and an idle power of 0.04W [88]. From these figures, the daily active usage time was calculated to be 165.5 minutes.

Feature Phones
It was estimated that feature phone ownership in 2025 would be 0.006 devices per person, resulting in a total stock of 0.41 million units [89]. The Nokia 105 was chosen as the representative model, a decision supported by reports that major UK mobile operators have seen a resurgence in non-smartphone sales, with the Nokia 105 being explicitly named as a popular model [90]. This device has a battery capacity of 2.96Wh [91]. Its charger draws 1.75W while charging and 0.075W when idle [92]. Using these data, the daily active usage time was calculated to be 112.8 minutes.

Tablets

The estimated stock of tablets in 2025 is 34.96 million units [93]. Apple is the dominant market leader with a 55% vendor market share [94]. Based on further market analysis, the iPad 10th generation was identified as the most-used model in 2025. Its battery capacity is 28.6Wh [96], and its charger has an active load of 12W and an idle load of 0.05W [95]. The corresponding daily active usage time was subsequently calculated to be 171.8 minutes.

Laptops

According to ECUK datasets, there are 31.86 million laptop units in domestic use [23]. Dell, a consistent top-three global vendor, was chosen as the representative brand. Its XPS 13 model line is widely regarded by UK tech reviewers as a benchmark for the premium ultraportable category, justifying its selection [97]. This laptop uses a 52Wh battery [98], and its charger has an active power draw of 42W and an idle draw of 0.5W [99]. Following the calculations, the daily active usage time was determined to be 219 minutes.

Handheld Consoles

The stock of handheld consoles is estimated to be 2.44 million units, a figure derived from an Ofcom report on UK households that own a Nintendo Switch. As the Switch represents the vast majority of the modern handheld market, this number was adopted for the entire category. The device's battery capacity is 16Wh [100]. The active charging power was determined to be 9.8W, representing the power drawn while the console is charging in sleep mode, a common scenario, and the idle standby power is 0.08W [101]. Based on these parameters and the calculation outlined in Section 4.2, the calculated daily usage time is 101.8 minutes.

5 Results and Analysis

This chapter presents the quantitative outputs from the bottom-up energy model. The analysis quantifies the annual operational energy consumption and corresponding carbon emissions for the 26 household electronic devices, grouped into four categories: ‘Kitchen and Cooking’, ‘Computing and Networking’, ‘Personal Use’, and ‘Entertainment and Media’. To benefit both policymakers and consumers, results are presented at both the national and per-device levels. To establish the model's credibility, its outputs are benchmarked against the 2023 Energy Consumption in the UK (ECUK) dataset, with a critical discussion of the variances observed. Finally, a Monte Carlo simulation and sensitivity analysis are conducted to explore the uncertainty inherent in the model's input parameters and their effect on the final estimations.

5.1 National Electricity Consumption and Operational Emissions

The analysis begins with a comprehensive overview of the national plug-load hierarchy for 2025. Figure 5.1 illustrates the annual electricity demand for all 26 appliances, while Figure 5.2 translates this energy consumption into operational carbon emissions.

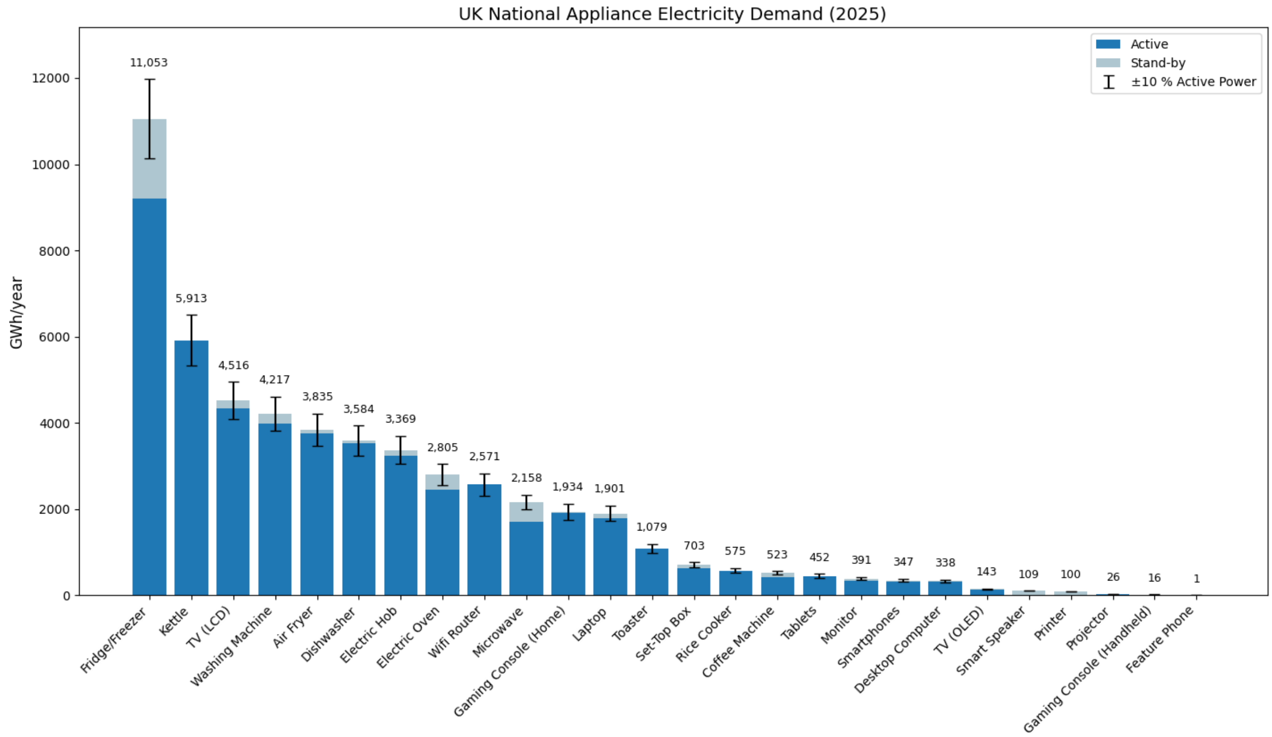

FIGURE 5.1: UK NATIONAL APPLIANCE ELECTRICITY DEMAND (2025)

A clear hierarchy emerges from the data. Cold appliances dominate the landscape: Fridge/ Freezers alone account for over 11 TWh per year, almost twice the electricity drawn by the next-largest load. They are followed by Kettles and LCD Televisions, each consuming just below 6 TWh and 5 TWh, confirming that both high-power kitchen heating devices and large-screen entertainment remain major contributors to residential demand. After this top tier, electricity use falls away quickly, illustrating a long tail of smaller loads. Standby energy is a visible but generally modest component for most products, with the notable exception of "always-on" devices like Wi-Fi routers.

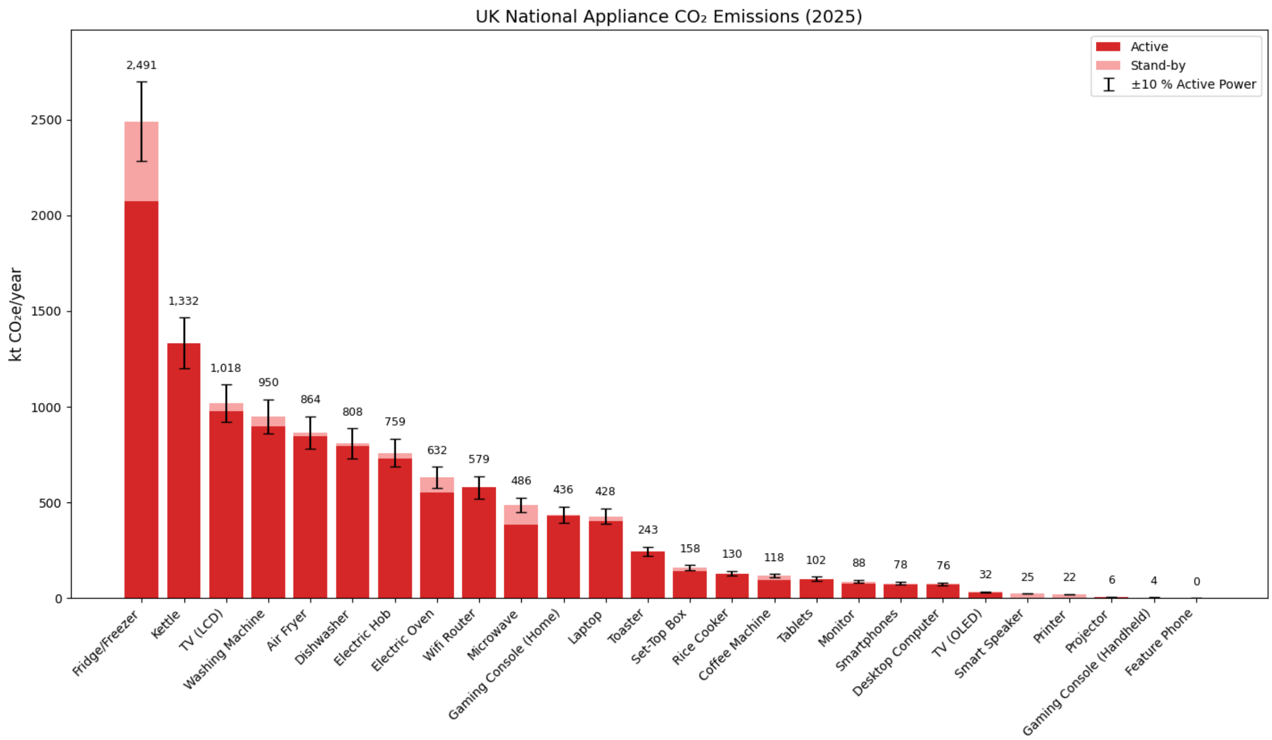

FIGURE 5.2: UK NATIONAL APPLIANCE CO_2E EMISSIONS (2025)

The operational emissions shown in Figure 5.2 directly mirror the energy consumption data, as they are calculated using a constant grid carbon factor. The Fridge/Freezer is the largest single emitter, responsible for approximately 2,491 kt of CO_2e per year. The Kettle (1,332 kt CO_2e/yr) and LCD Television (1,018 kt CO_2e/yr) follow, reinforcing their status as high-priority targets for national energy policy.

5.2 Category Analysis: National Consumption and Emissions

The following sections present the aggregated national impact of household appliances, broken down by functional category. This national perspective is crucial for identifying the devices that contribute most significantly to the UK's overall residential energy demand and carbon footprint.

5.2.1 Kitchen and Cooking Appliances

The simulation, which scales per-device data by official ownership figures, showcases the significant impact of kitchen appliances on the UK energy system. The primary results are summarised in Table 5.1, and the bar charts in Figure 5.3 and Figure 5.4 visualise these national impacts.

Device	Annual Device-Level Consumption (kWh/yr)	National Annual Consumption (GWh/yr)	National Annual Emissions (kt CO_2e/yr)	Annual Device-Level Emissions (kg CO_2e/yr)
Fridge/Freezer	525.6	11053.4	2490.9	118.4
Kettle	219.0	5913.0	1332.5	49.4
Washing Machine	153.3	4216.8	950.2	34.6
Air Fryer	232.4	3835.1	864.2	52.4
Dishwasher	252.4	3584.4	807.8	56.9
Electric Hob	227.6	3369.0	759.2	51.3
Electric Oven	134.2	2804.7	632.0	30.2
Microwave	84.3	2158.2	486.3	19.0
Toaster	49.3	1079.1	243.2	11.1
Rice Cooker	127.7	574.9	129.5	28.8
Coffee Machine	32.3	523.0	117.8	7.3

5.1 SUMMARY OF NATIONAL AND DEVICE-LEVEL ENERGY CONSUMPTION AND EMISSIONS FOR KITCHEN AND COOKING APPLIANCES

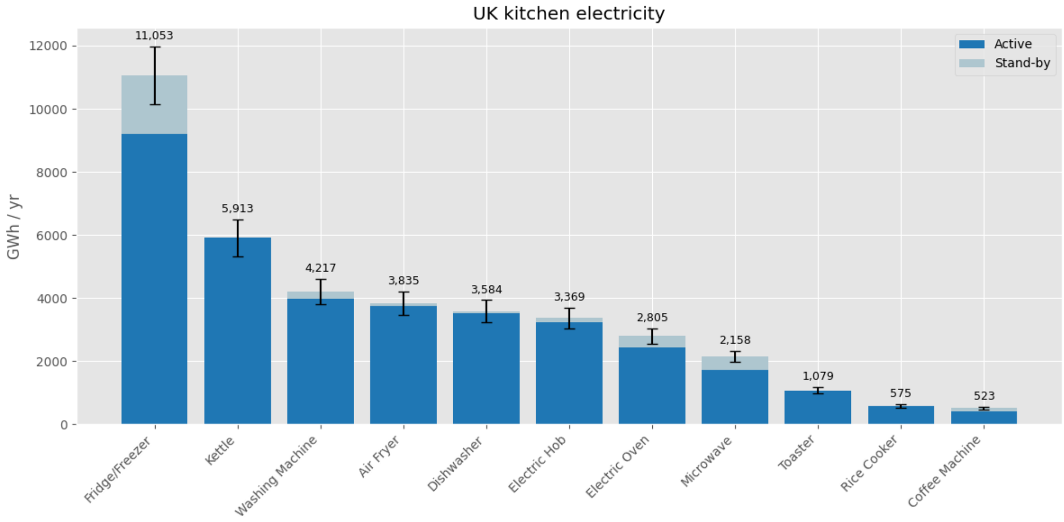

FIGURE 5.3: NATIONAL ANNUAL ELECTRICITY CONSUMPTION FOR UK KITCHEN APPLIANCES (GWH/YR)

As illustrated in Figure 5.3, the Fridge-Freezer is the most energy-intensive device, consuming 11,053 GWh/yr. Its significant standby power consumption, which accounts for the periods when the compressor is off, contributes substantially to this total. The Kettle follows at 5,913 GWh/yr; its high ranking is driven by its powerful 3000W heating element and near-universal ownership. A middle tier of appliances, from the Washing Machine (4,217 GWh/yr) to the Electric Oven (2,805 GWh/yr), also represents a major portion of national demand. The high standing of devices like the Washing Machine and Kettle is strongly linked to their high ownership rates, while the Fridge-Freezer's dominance is a result of its continuous 24/7 operation.

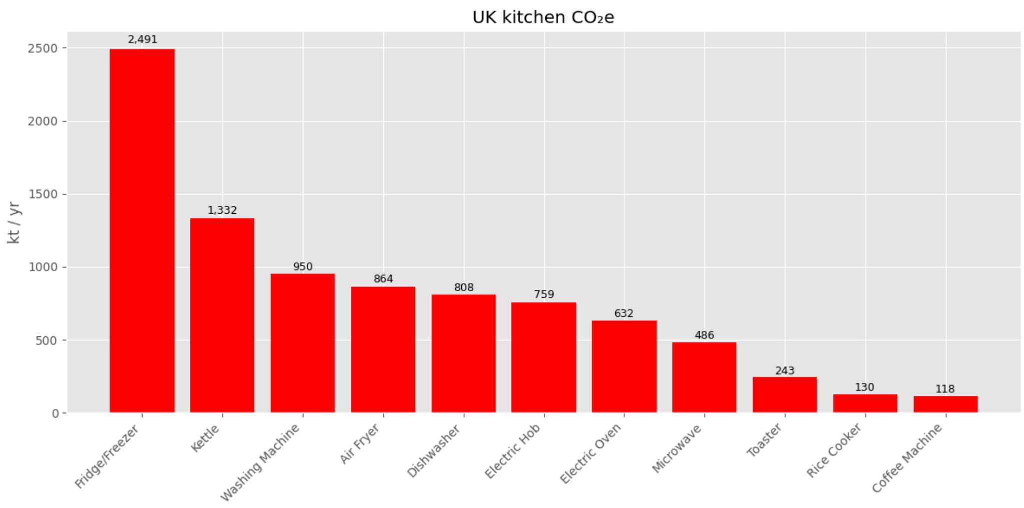

FIGURE 5.4: NATIONAL ANNUAL CO2 E EMISSIONS FOR UK KITCHEN APPLIANCES (KT/YR)

The operational carbon emissions, shown in Figure 5.4, directly correlate with electricity consumption. The Fridge/Freezer is the largest emitter at 2,491 kt CO_2e/yr, followed by the Kettle at 1,332 kt CO_2e/yr. These two appliances alone account for a substantial portion of the kitchen's carbon footprint. The subsequent ranking mirrors that of energy consumption, reinforcing that the primary drivers of emissions are a combination of continuous operation, high power intensity, and widespread ownership.

5.2.2 Computing Networking Devices

The analysis of computing and networking devices reveals a distinct pattern where continuous, low-power operation can rival intermittent, high-power use at a national scale. The results are summarised in Table 5.2.

Device	Annual Device-Level Consumption (kWh/yr)	National Annual Consumption (GWh/yr)	National Annual Emissions (kt CO_2e/yr)	Annual Device-Level Emissions (kg CO_2e/yr)
Wifi Router	95.3	2571.4	579.5	21.5
Laptop	59.7	1901.2	428.4	13.4
Monitor	20.3	390.6	88.0	4.6
Desktop Computer	87.9	337.6	76.1	19.8
Printer	12.3	99.6	22.5	2.8
Projector	43.6	26.2	5.9	9.8

5.2 SUMMARY OF NATIONAL AND DEVICE-LEVEL ENERGY CONSUMPTION AND EMISSIONS FOR COMPUTING AND NETWORKING DEVICES

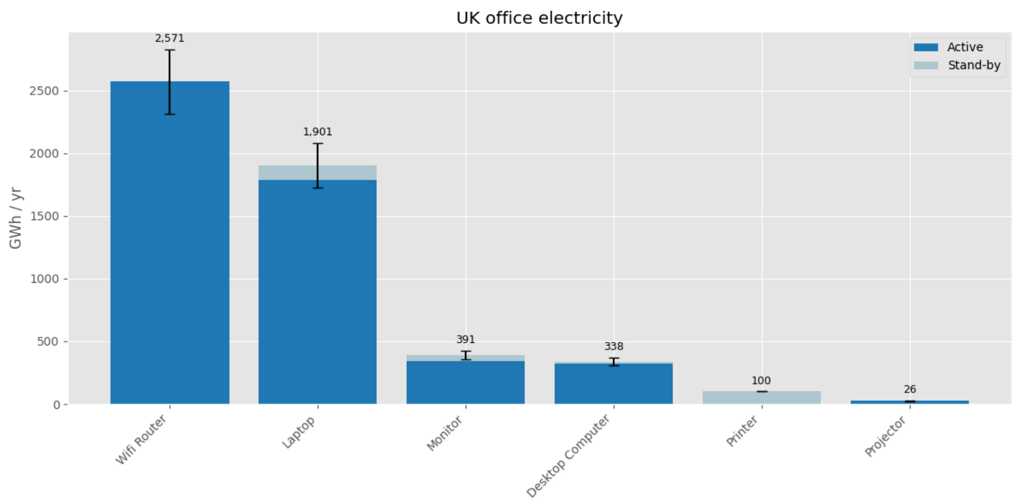

FIGURE 5.5: NATIONAL ANNUAL ELECTRICITY CONSUMPTION FOR UK COMPUTING AND NETWORKING DEVICES (GWH/YR)

As shown in Figure 5.5, the Wifi Router emerges as the largest national energy consumer in this category at 2,571 GWh/yr, a finding that underscores the impact of 'always-on' devices. Despite its low power draw, its 24/7 operational time, multiplied across nearly 27 million households, results in a substantial national footprint. Laptops follow at 1,901 GWh/yr, their impact driven by the highest ownership in this category (~32 million units). In contrast, Desktop Computers, despite having a higher power draw, have a much lower national impact (338 GWh/yr) due to their significantly smaller ownership base. Printers and projectors have negligible electricity consumption due to the very low usage time for printers and the very small stock of ~600,000 units in the UK.

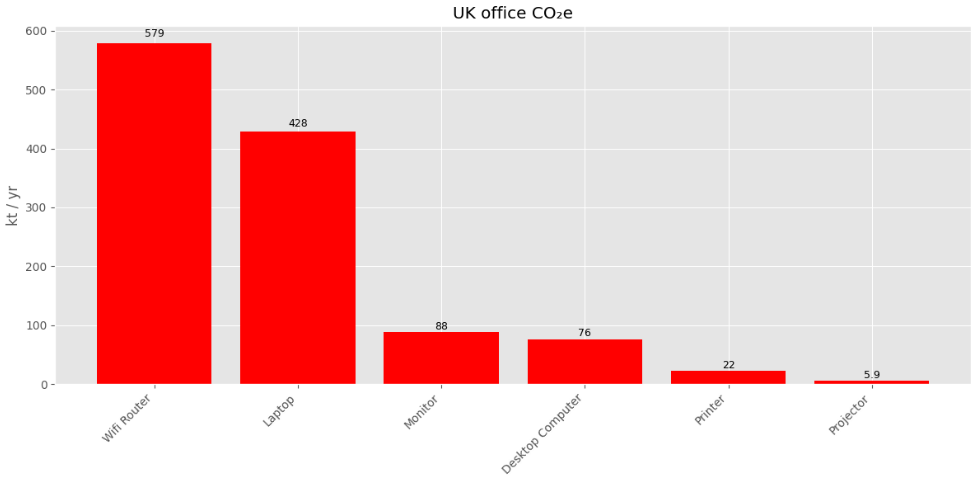

FIGURE 5.6: NATIONAL ANNUAL CO_2E EMISSIONS FOR UK COMPUTING AND NETWORKING DEVICES (KT/YR)

As shown in Figure 5.6, the operational emissions directly mirror the energy consumption hierarchy. The Wifi Router is the largest emitter at 579 kt CO_2e/yr, followed by the Laptop at 428 kt CO_2e/yr. This highlights that, from a policy perspective, targeting the baseline efficiency of ubiquitous, continuously operating devices can be as critical as addressing high-power, active-use appliances.

5.2.3 Personal Use Devices

This category primarily includes portable, battery-operated devices such as smartphones and tablets, which share a distinct calculation methodology based on charging cycles, and mains-powered smart speakers.

Device	Annual Device-Level Consumption (kWh/yr)	National Annual Consumption (GWh/yr)	National Annual Emissions (kt CO_2e/yr)	Annual Device-Level Emissions (kg CO_2e/yr)
Tablets	12.9	451.9	101.8	2.9
Smartphones	5.3	347.0	78.2	1.2
Smart Speaker	11.6	109.0	24.6	2.6
Feature Phone	1.8	0.7	0.2	0.4

5.3: SUMMARY OF NATIONAL AND DEVICE-LEVEL ENERGY CONSUMPTION AND EMISSIONS FOR PERSONAL USE DEVICES

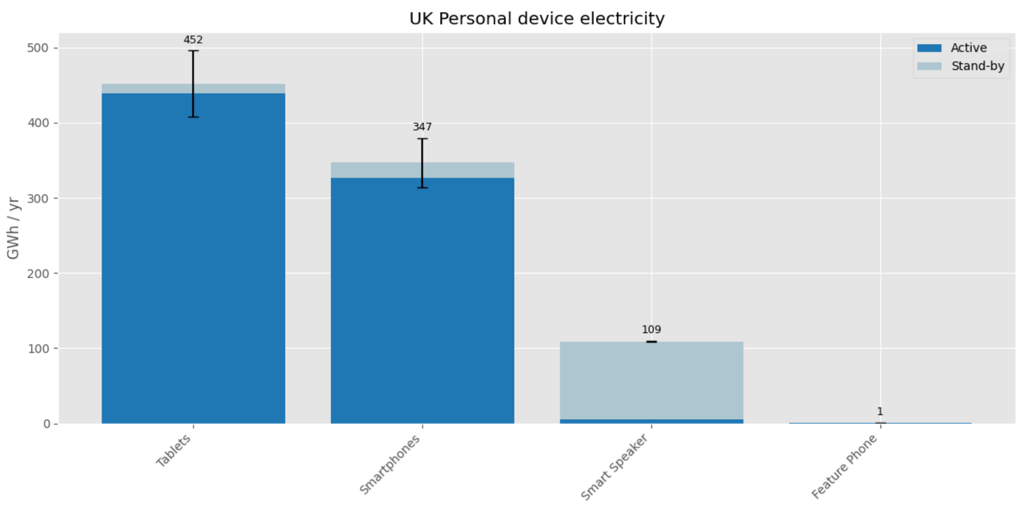

FIGURE 5.7: NATIONAL ANNUAL ELECTRICITY CONSUMPTION FOR UK PERSONAL USE DEVICES (GWH/YR)

At the national level, Tablets (452 GWh/yr) and Smartphones (347 GWh/yr) are the most significant energy consumers. Their impact is a direct result of their vast ownership numbers, 35 million and 65 million, respectively. A key finding for this category is the consumption profile of the Smart Speaker; although its active use is minimal, its high standby power draw while waiting for a user command makes up the vast majority of its 109 GWh/yr national consumption.

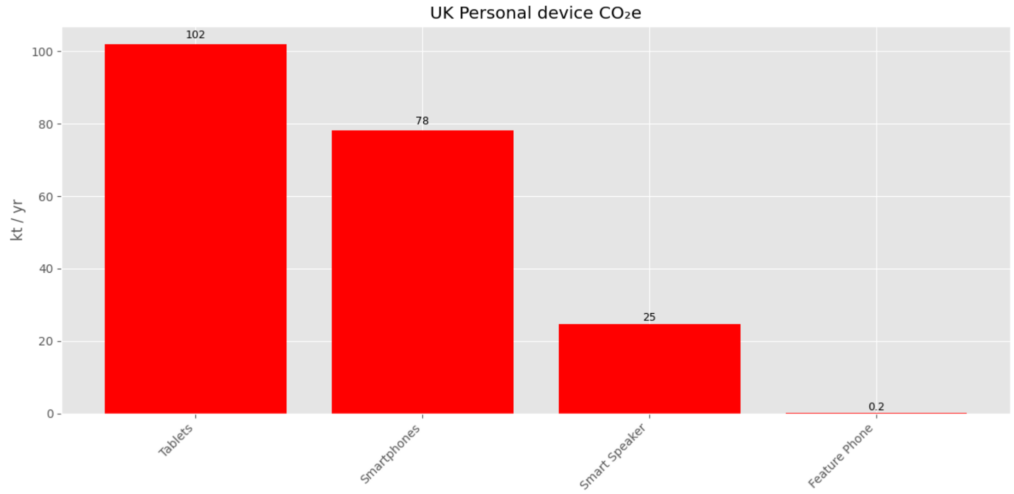

FIGURE 5.8: NATIONAL ANNUAL CO_2E EMISSIONS FOR UK PERSONAL USE DEVICES (KT/YR)

The emissions ranking in Figure 5.8 directly follows the consumption data, with Tablets (102 kt CO_2e/yr) and Smartphones (78 kt CO_2e/yr) being the largest contributors. The results underscore the collective carbon footprint of widely adopted personal electronics.

5.2.4 Entertainment and Media Devices

The Entertainment and Media category is dominated by devices with high power consumption and long daily usage times, particularly televisions, which are a cornerstone of household leisure. However, the findings for this category, especially for LCD TVs, represent a significant anomaly in the dataset and should be interpreted with caution, as will be explored in the validation section.

Device	Annual Device-Level Consumption (kWh/yr)	National Annual Consumption (GWh/yr)	National Annual Emissions (kt CO_2e/yr)	Annual Device-Level Emissions (kg CO_2e/yr)
TV (LCD)	86.3	4515.6	1017.6	19.5
Gaming Console (Home)	198.0	1934.3	435.9	44.6
Set-Top Box	27.0	703.1	158.5	6.1
TV (OLED)	136.6	143.4	32.3	30.8
Gaming Console (Handheld)	6.7	16.4	3.7	1.5

5.4 SUMMARY OF NATIONAL AND DEVICE-LEVEL ENERGY CONSUMPTION AND EMISSIONS FOR ENTERTAINMENT AND MEDIA DEVICES

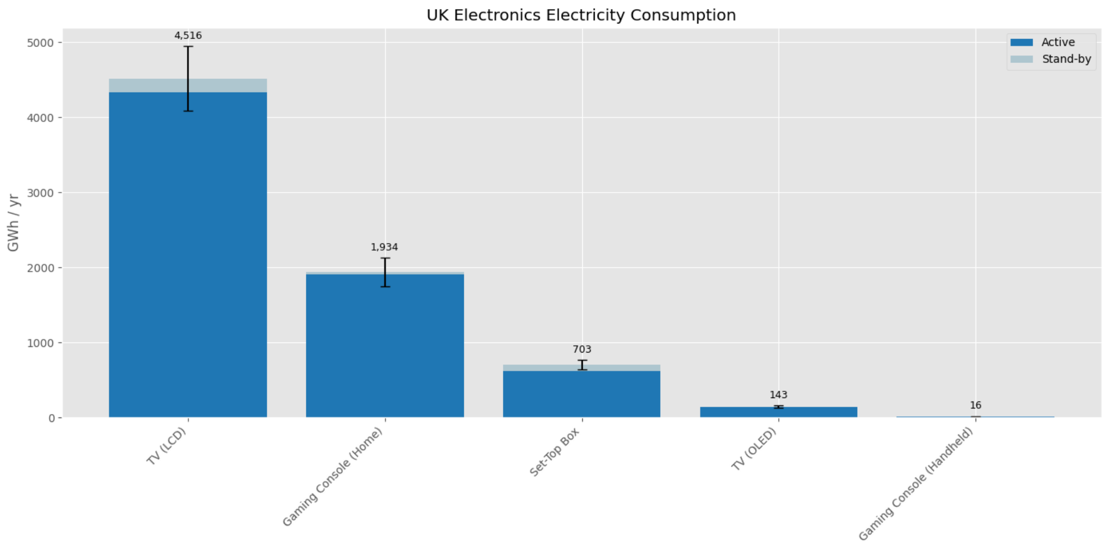

FIGURE 5.9: NATIONAL ANNUAL ELECTRICITY CONSUMPTION FOR UK ENTERTAINMENT AND MEDIA DEVICES (GWH/YR)

At the national scale, LCD televisions dominate the electronics load, drawing an estimated 4.5 TWh/yr. That value is driven by the sheer scale of ownership, just over 52 million sets split almost evenly between primary and secondary TVs, and by their long daily runtimes. Even the high-power home gaming console, the second-ranked device, uses only 1.9 TWh/yr, roughly one-third of the LCD total. Set-top boxes follow at 0.7 TWh/yr, while OLED TVs and handheld consoles contribute only 0.14 TWh and 0.02 TWh, respectively, reflecting their much lower market penetration.

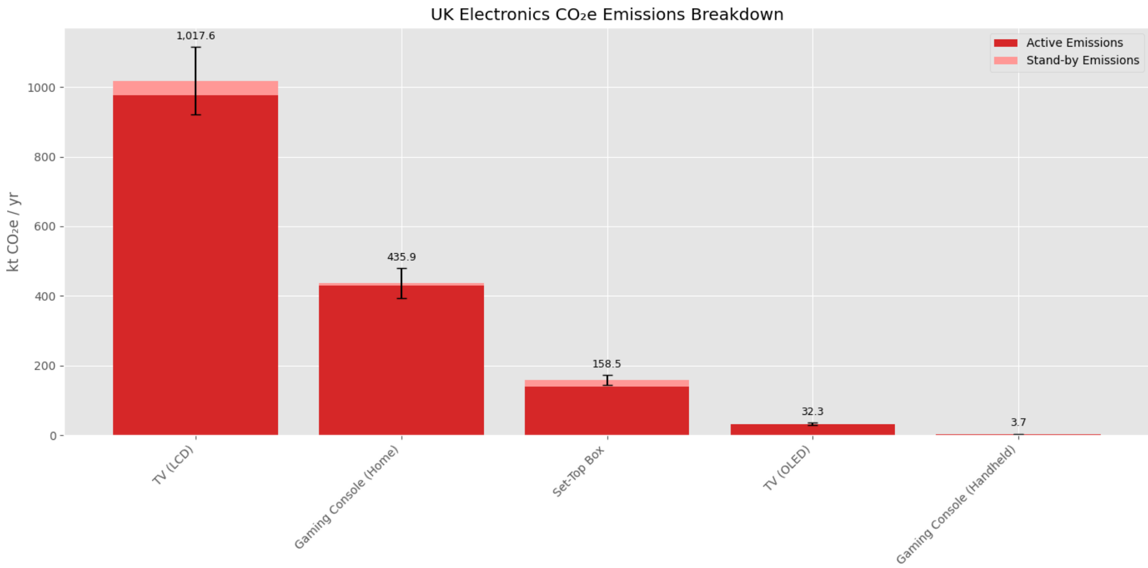

FIGURE 5.10: NATIONAL ANNUAL CO_2E EMISSIONS FOR UK ENTERTAINMENT AND MEDIA DEVICES (KT/YR)

The emissions picture in Figure 5.10 mirrors this order: LCD TVs account for about 1.02 Mt CO_2e/yr, more than three times the 0.44 Mt CO_2e attributed to home gaming consoles. Set-top boxes add a further 0.16 Mt CO_2e, whereas OLED TVs and handheld consoles together contribute well under 0.04 Mt CO_2e. Although these numbers point to televisions as the prime efficiency target, the uncertainty bands on active power underline the need for empirical validation, particularly of average screen size, brightness settings and viewing hours, before translating the headline figures into policy actions.

5.3 Deconstructing the national impact: Device-specific annual energy consumption

While national figures are essential for policymakers as they incorporate ownership data, analysing device-level consumption and emission rates reveals the energy intensity of individual devices. This perspective provides valuable data for residents seeking to understand and manage their energy use and associated costs.

5.3.1 Kitchen and Cooking Appliances

At the device-level, the energy intensity of individual appliances becomes clearer, providing a different perspective from the national overview.

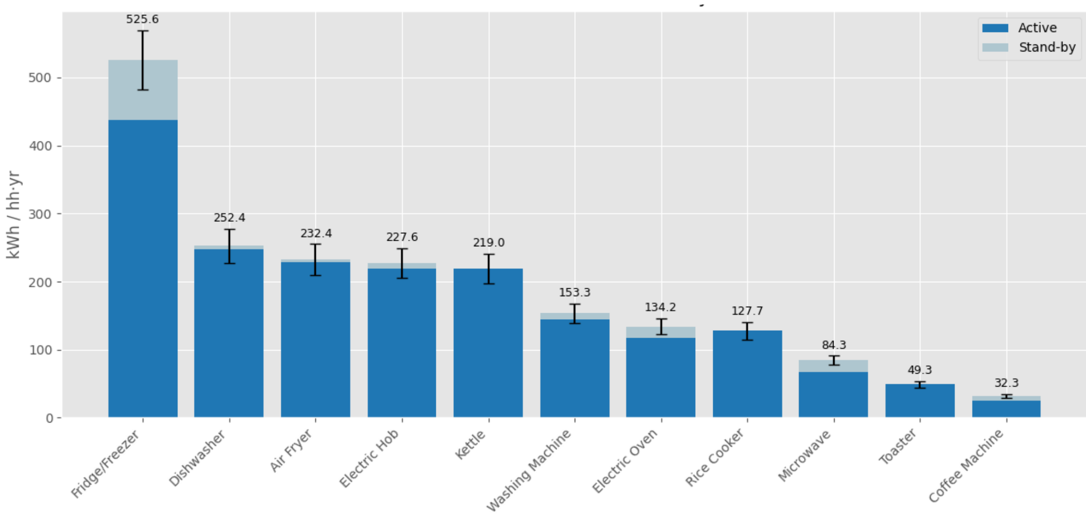

FIGURE 5.11: PER-HOUSEHOLD ANNUAL ENERGY CONSUMPTION FOR KITCHEN AND COOKING APPLIANCES (KWH/YR)

As shown in Figure 5.11, the Fridge/Freezer remains the most energy-intensive appliance within a single home, consuming 525.6 kWh per year. However, the ranking of other appliances shifts significantly compared to the national picture; the Dishwasher (252.4 kWh/yr) and Air Fryer (232.4 kWh/yr) demonstrate higher device-level intensity than the Kettle (219.0 kWh/yr). This highlights a critical distinction: while the kettle's national impact is enormous due to its massive stock, other devices that run for longer durations are more demanding within a single home. This distinction is vital for targeting different types of policy interventions. National-level issues like the kettle suggest a need for technological standards (Ecodesign), whereas high-intensity household devices like the dishwasher point towards the importance of influencing user behaviour, for example, 'Energy Labelling Awareness Campaigns' that highlight high running costs or public information campaigns (e.g., "run only full loads").

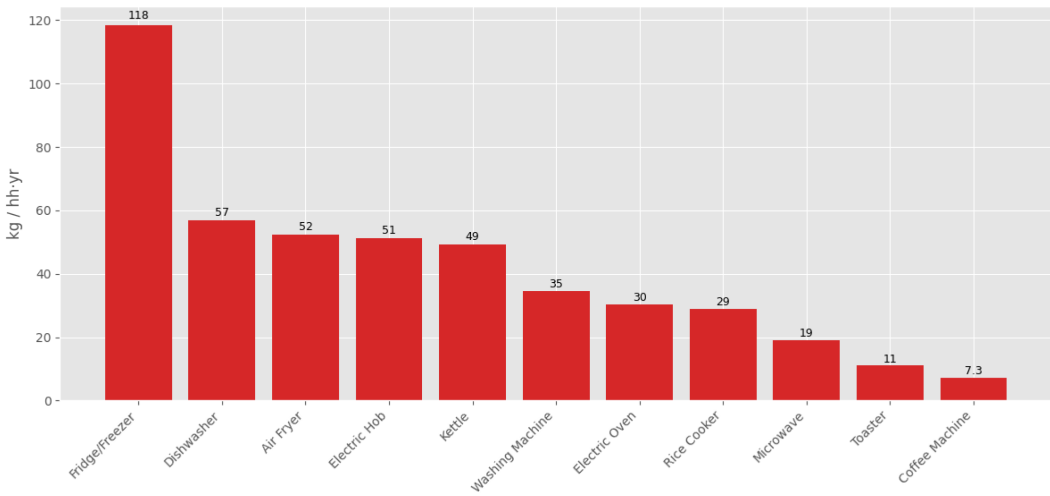

FIGURE 5.12: PER-HOUSEHOLD ANNUAL CO_2E EMISSIONS FOR KITCHEN AND COOKING APPLIANCES (KG/YR)

The device-level carbon footprint, illustrated in Figure 5.12, follows the same pattern as energy use. The Fridge/Freezer generates the most emissions per household at 118 kg CO_2e/yr. The Dishwasher (57 kg), Air Fryer (52 kg), and Electric Hob (51 kg) form a clear second tier of high-intensity emitters, all surpassing the Kettle (49 kg). This analysis reinforces that for an individual household seeking to reduce its carbon footprint, managing the frequency and cycle choices of high-intensity appliances like dishwashers can be as impactful as behavioural changes related to kettle use, such as the optimal use of boiling water by only filling it to the required level.

5.3.2 Computing and Networking Devices

At the household level, the energy intensity rankings for computing devices shift, revealing a different story from the national picture.

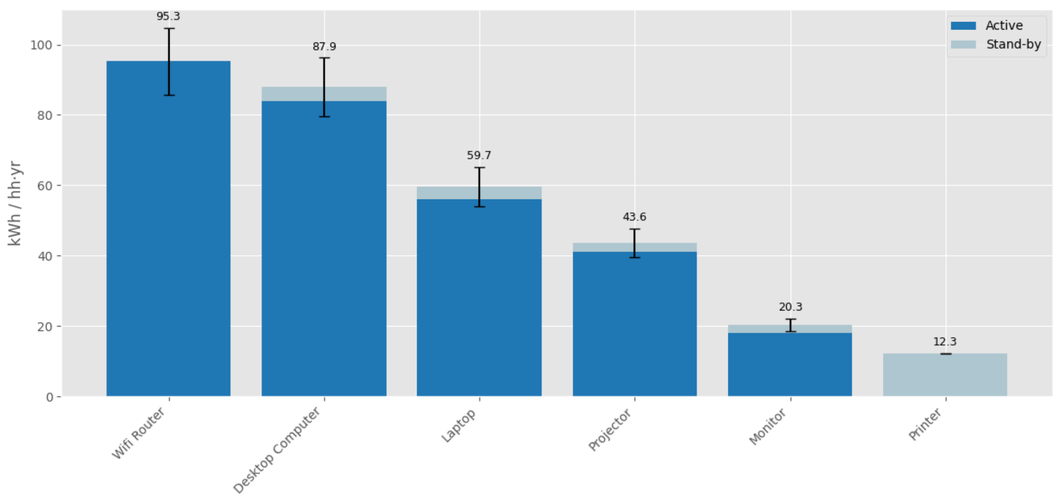

FIGURE 5.13: PER-HOUSEHOLD ANNUAL ENERGY CONSUMPTION FOR COMPUTING AND NETWORKING DEVICES (KWH/YR)

As seen in Figure 5.13, the Wi-Fi router remains the most energy-intensive device within a single household at 95.3 kWh/yr, due to its continuous operation. It is closely followed by the Desktop Computer at 87.9 kWh/yr, whose high power draw during active use results in significant consumption. The Laptop, despite its large national impact due to high ownership, consumes considerably less per household (59.7 kWh/yr). Projectors (43.6 kWh/yr), who were at the bottom initially, have surpassed monitors and printers, since their low number of units was the only reason they were that low. A notable finding is the Printer, where standby power constitutes the vast majority of its annual consumption, as its active use is extremely brief (0.15 minutes/day). This demonstrates that for many modern electronics, standby and idle loads are a primary driver of household energy costs, as well as how national standards do not help home users know which of their electronics they should focus on for energy saving due to their intensity.

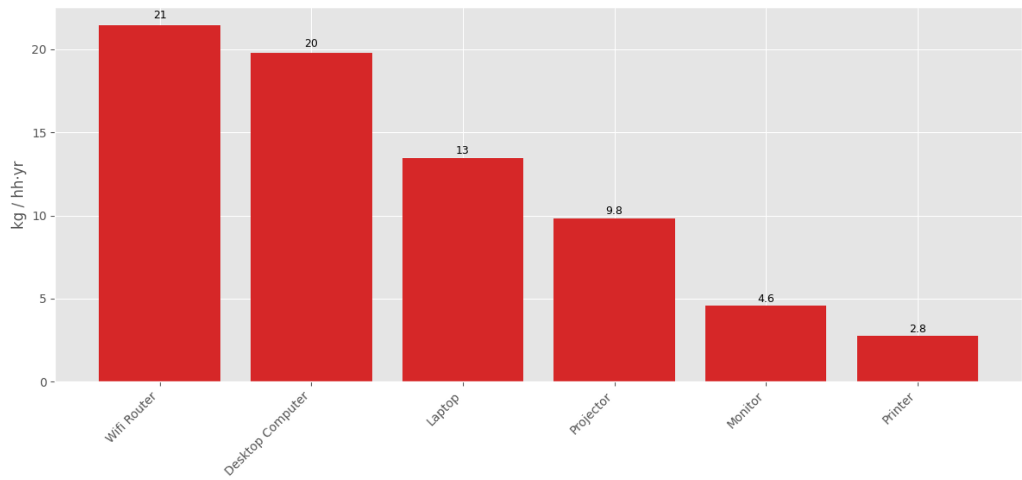

FIGURE 5.14: PER-HOUSEHOLD ANNUAL CO_2E EMISSIONS FOR COMPUTING AND NETWORKING DEVICES (KG/YR)

The device-level carbon footprint, shown in Figure 5.14, directly reflects the energy consumption data. The Wifi Router (21.5 kg CO_2e/yr) and Desktop Computer (19.8 kg CO_2e/yr) are the most significant emitters in a typical home. This analysis suggests that for individuals, strategies such as using smart plugs to power down devices completely when not in use could yield tangible reductions in both energy bills and carbon emissions.

5.3.3 Personal Use Devices

The per-household analysis of personal devices reveals a different energy intensity hierarchy compared to the national figures, primarily driven by standby power consumption

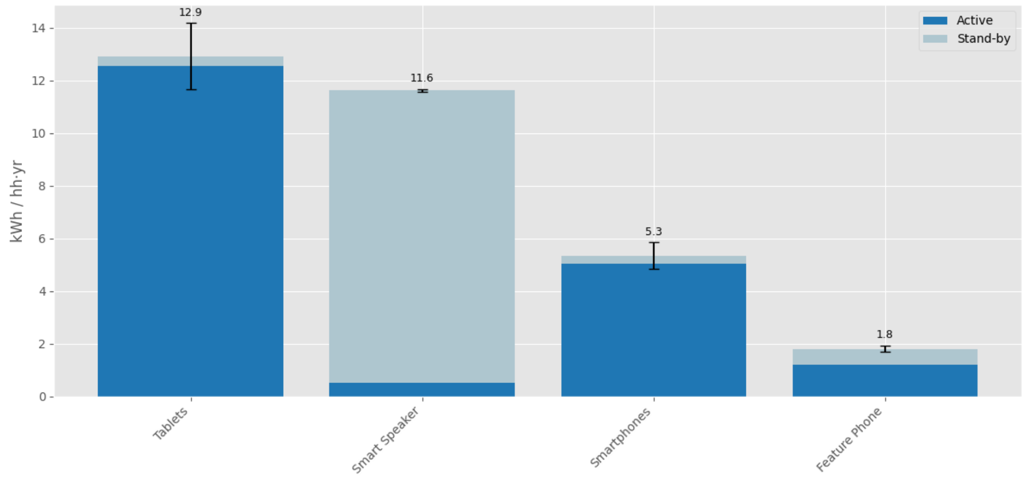

FIGURE 5.15: PER-HOUSEHOLD ANNUAL ENERGY CONSUMPTION FOR PERSONAL USE DEVICES (KWH/YR)

As shown in Figure 5.15, the Tablet is the most energy-intensive device per household at 12.9 kWh/yr. Notably, the Smart Speaker (11.6 kWh/yr) surpasses the Smartphone (5.3 kWh/yr) in household consumption. This is a critical insight that highlights the difference in operation between these devices. As a battery-powered device, a smartphone is not always drawing power from the mains, whereas the Smart Speaker is an 'always-on' device whose energy use is almost entirely composed of its standby power draw while waiting for a command. While smartphones dominate the national consumption due to 95% population ownership, the smart speaker is inherently more energy-intensive per device within the home. This reinforces the finding from the computing category, particularly with the Printer, that standby loads are a significant and often overlooked component of domestic energy use.

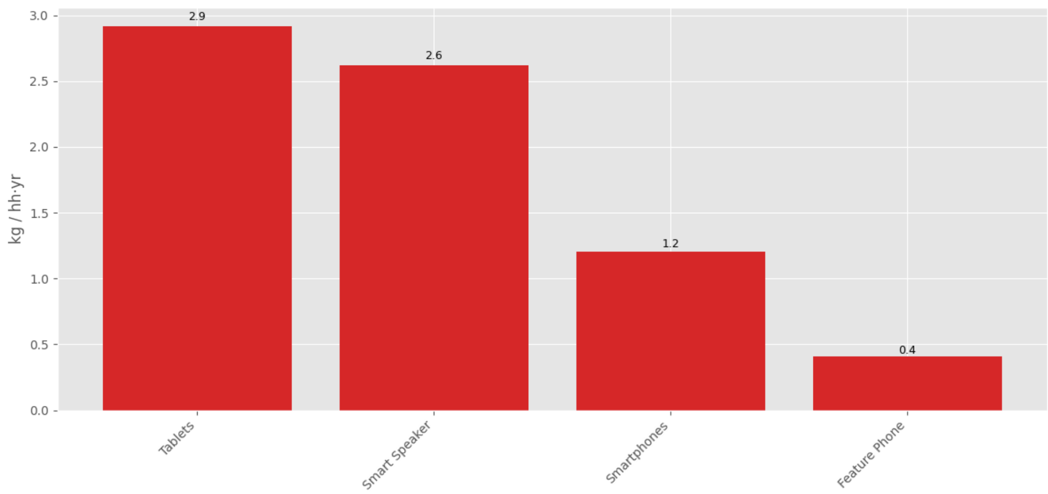

FIGURE 5.16: PER-HOUSEHOLD ANNUAL CO2 E EMISSIONS FOR PERSONAL USE DEVICES (KG/YR)

The device-level emissions data in Figure 5.16 mirrors this trend, with Tablets (2.9 kg CO_2e/yr) and Smart Speakers (2.6 kg CO_2e/yr) having the largest carbon footprints within a home. This suggests that for consumers, awareness of standby consumption is as important as managing active charging cycles for reducing personal emissions. Actionable steps could include unplugging devices with high standby loads, such as a smart speaker, when leaving the house for extended periods.

5.3.4 Entertainment and Media Devices

While LCD TVs dominate the national figures due to sheer volume, the per-household analysis reveals that other devices are far more energy-intensive during use.

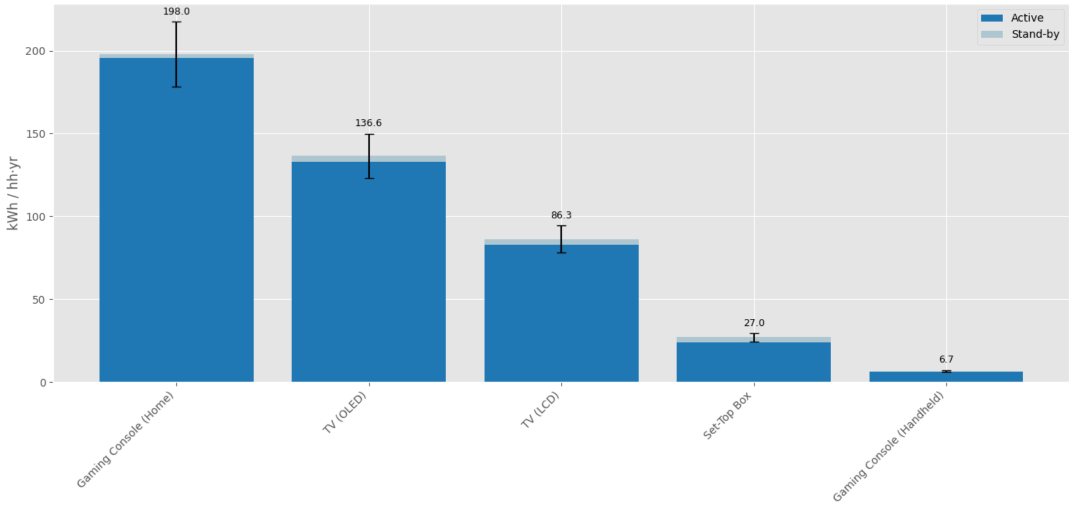

FIGURE 5.17: PER-HOUSEHOLD ANNUAL ENERGY CONSUMPTION FOR ENTERTAINMENT AND MEDIA DEVICES (KWH/YR)

Figure 5.17 shows that a home gaming console is the single most energy-intensive electronic device in an average household, drawing about 198 kWh/yr. Its demand comfortably exceeds that of televisions: OLED sets average 137 kWh/yr and LCD sets 86 kWh/yr. The disparity is driven by the console's very high active-mode power during gameplay, well above a typical TV's draw, and it compounds when the console and screen operate together, meaning a couple of hours of gaming can outstrip the electricity used by an entire evening of television viewing. Lower down the ranking, set-top boxes consume roughly 27 kWh/yr, while handheld consoles barely register at 7 kWh/yr. Standby consumption is almost invisible for every device, and the ±10% error bars around active power confirm that a reasonable uncertainty in wattage does not disturb the overall ordering.

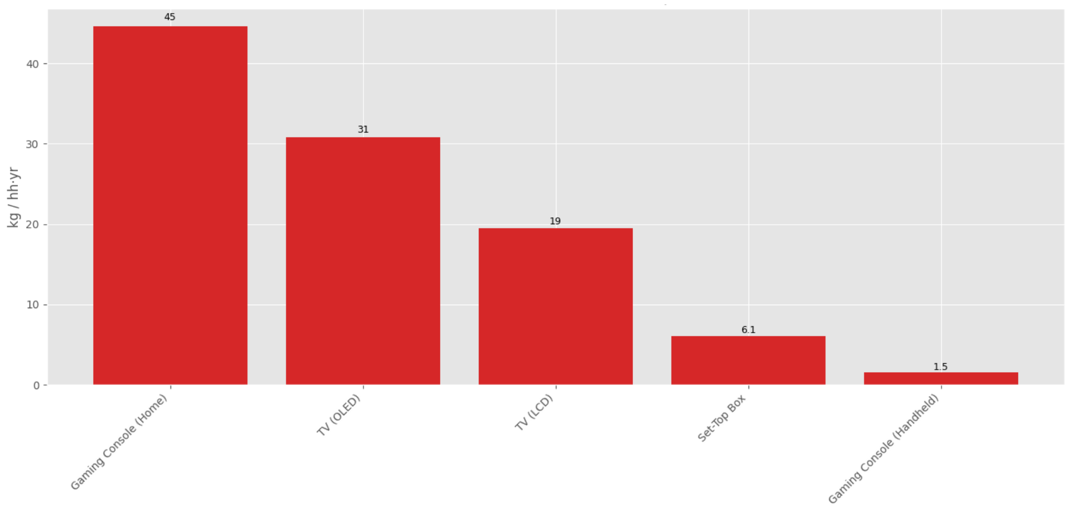

FIGURE 5.18: PER-HOUSEHOLD ANNUAL CO_2E EMISSIONS FOR ENTERTAINMENT AND MEDIA DEVICES (KG/YR)

Per-household emissions follow the same pattern as energy use. Home gaming consoles top the chart, adding roughly 45 kg CO_2e per year, while an OLED and an LCD television account for about 31 kg CO_2e and 19 kg CO_2e, respectively. By contrast, a set-top box contributes only 6 kg CO_2e, and a handheld console scarcely 1 kg CO_2e. For households aiming to reduce their carbon footprint, moderating gaming hours, selecting energy-saving console settings, or shutting the device down fully instead of leaving it in rest mode offers a far greater benefit than incremental changes to smaller electronics.

5.4 Model validation against ECUK Benchmark

To establish the model's credibility, its outputs for national energy consumption were benchmarked against the ECUK 2023 dataset. On the comparative plot, the percentage difference is colour-coded for clarity: green indicates a variance of less than 10%, orange indicates a variance of less than 25%, and red indicates a variance greater than 25%. It should be noted that the 'Personal Use' device category is excluded from this validation, as ECUK does not provide comparable data for these specific appliances.

5.4.1 Kitchen appliances

Device	Model (GWh/yr)	ECUK (GWh/yr)	Percentage Difference
Fridge/Freezer	11,053	6,019	83.6%
Kettle	5,913	4,843	22.1%
Dishwasher	3,584	3,502	2.4%
Electric Hob	3,369	2,657	26.8%
Microwave	2,158	2,507	-13.9%
Washing Machine	4,217	6,773	-37.7%
Electric Oven	2,805	2,008	39.7%

5.5 MODEL VALIDATION: COMPARISON OF NATIONAL ELECTRICITY CONSUMPTION FOR KITCHEN APPLIANCES AGAINST ECUK 2023 BENCHMARK

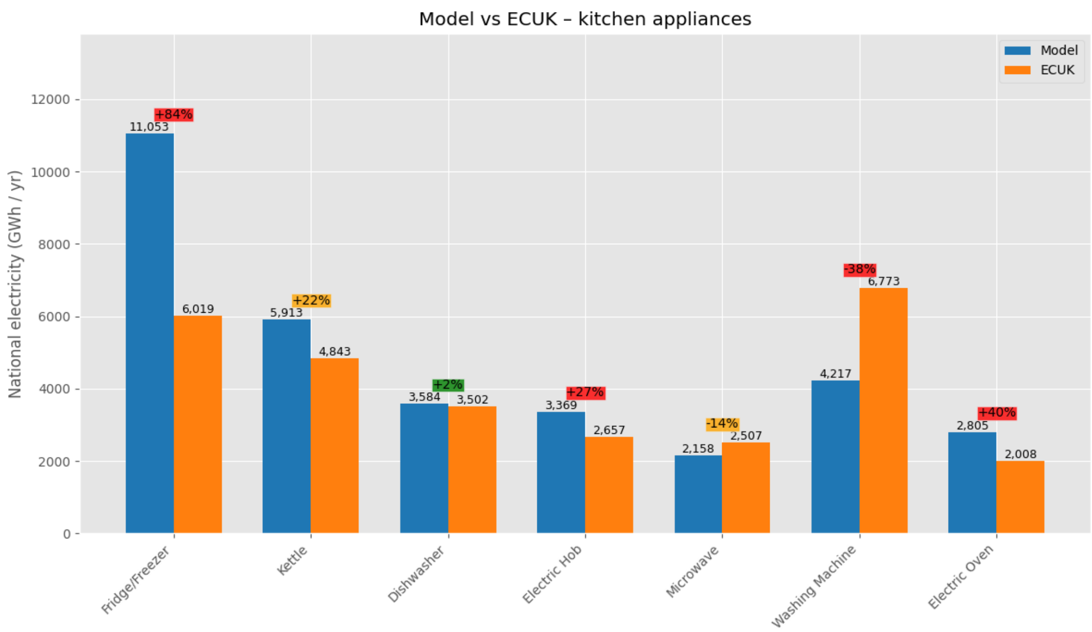

FIGURE 5.19: MODEL VALIDATION: COMPARISON OF NATIONAL ELECTRICITY CONSUMPTION FOR KITCHEN APPLIANCES

The validation exercise reveals a strong alignment for Dishwashers (+2.4%) but notable discrepancies for other appliances. These variances are not necessarily indicative of model error, but rather highlight the fundamental differences between this study's bottom-up methodology and the top-down approach used for the ECUK data. As discussed in the literature review, the ECUK figures are derived from the Building Research Establishment Domestic Energy Model (BREDEM), which relies on standardised assumptions about occupancy and building characteristics rather than device-specific parameters.

With this context, the discrepancies can be interpreted:

- **Fridge/Freezers (+83.6%):** This significant divergence is the clearest example of the methodological differences. The BREDEM model uses a formula based on floor area and occupancy to derive a single figure for all appliance consumption. 1 In contrast, this study's bottom-up model explicitly accounts for the 24/7 power draw (both active and standby) of the 21 million fridges/freezers in the UK. The BREDEM approach does not capture this device-specific, continuous load, which is a likely cause for its lower estimate.

- **Washing Machine (-37.7%):** The under-prediction may be due to this model's use of an average usage profile. The ECUK's standardised assumptions may account for more energy-intensive behaviours, such as a higher frequency of hot water cycles, leading to a higher national estimate.

- **Cooking Appliances (Oven +39.7%, Hob +26.8%):** This model's higher figures could be a result of using more current usage data that reflects modern cooking habits, which may differ from the fixed, older profiles embedded within the BREDEM framework, some of which date back to 2007.

Ultimately, this comparison serves not just to validate the model but to demonstrate the analytical value of a granular, bottom-up approach in providing an alternative, and potentially more current, perspective on appliance energy consumption.

5.4.2 Computing and Networking Devices

Device	Model (GWh/yr)	ECUK (GWh/yr)	Percentage Difference
Desktop Computer	338	668	-49.5%
Laptop	1,901	1,982	-4.1%
Monitor	391	353	10.6%
Printer	100	69	44.4%

5.6 MODEL VALIDATION: COMPARISON OF NATIONAL ELECTRICITY CONSUMPTION FOR OFFICE EQUIPMENT AGAINST ECUK 2023 BENCHMARK

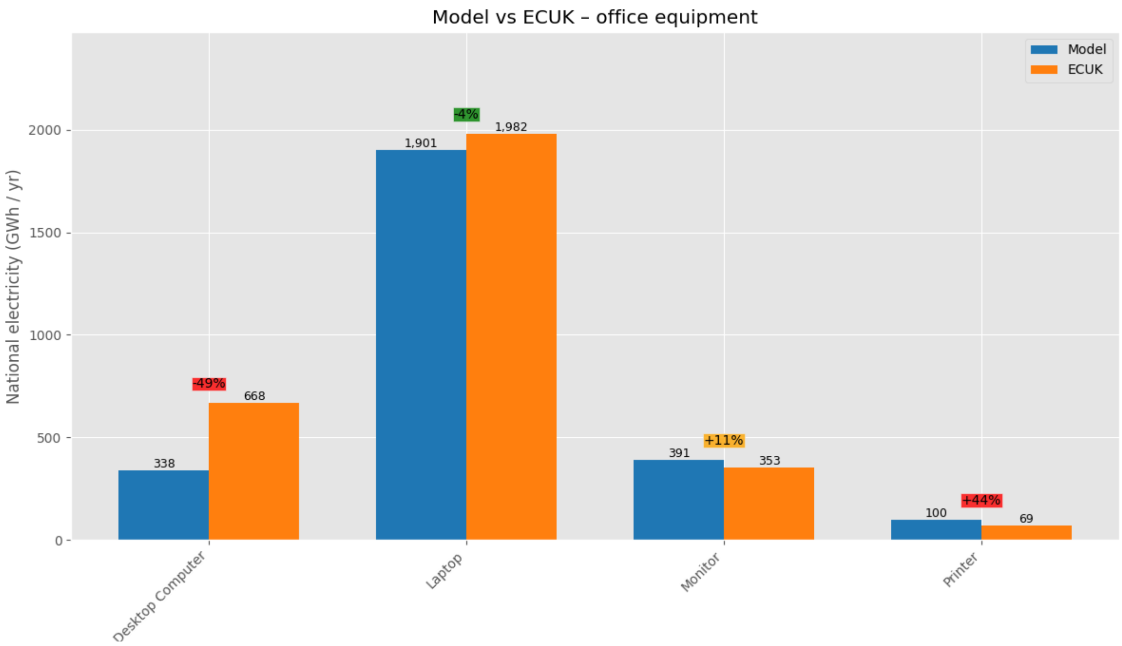

FIGURE 5.20: MODEL VALIDATION: COMPARISON OF NATIONAL ELECTRICITY CONSUMPTION FOR OFFICE EQUIPMENT

The model shows excellent alignment for Laptops, with only a -4.1% variance, suggesting the input parameters for this device are robust. However, there are significant discrepancies for other devices, which this model's assumptions can explain:

- **Desktop Computer (-49.5%):** The model significantly under-predicts desktop consumption. A key assumption in this model is the focus on active power, which does not fully capture the substantial energy consumed during standby or sleep modes, a common state for desktops. Another assumption that may have led to the underprediction was the assumption to mitigate gaming computers. As gaming computers normally run intensive tasks, they require higher amounts of power. These exclusions are a likely cause of the under-prediction.

- **Monitor (+10.6%):** The moderate over-prediction could stem from the assumption that monitors are active for the full duration of computer use, whereas in reality, they often have more aggressive, independent power-saving features.

- **Printer (+44.4%):** While the percentage difference is large, the absolute energy values are small (100 GWh vs 69 GWh), making the variance less impactful on the national total. The difference likely arises from this model's assumption of a specific printing frequency (30 pages per month) and standby power draw, which may differ from the assumptions in the BREDEM framework.

5.4.3 Entertainment and Media

The validation for entertainment and media devices reveals the most significant discrepancies between this model and the ECUK benchmark, particularly for televisions.

Device	Model (GWh/yr)	ECUK (GWh/yr)	Percentage Difference
Gaming Console (Home)	1,934	1,677	15.3%
TV (LCD)	4,516	1,252	260.7%
TV (OLED)	143	56	156.1%
Set-Top Box	703	1,134	-38.0%

5.7 MODEL VALIDATION: COMPARISON OF NATIONAL ELECTRICITY CONSUMPTION FOR ENTERTAINMENT AND MEDIA DEVICES AGAINST ECUK 2023 BENCHMARK

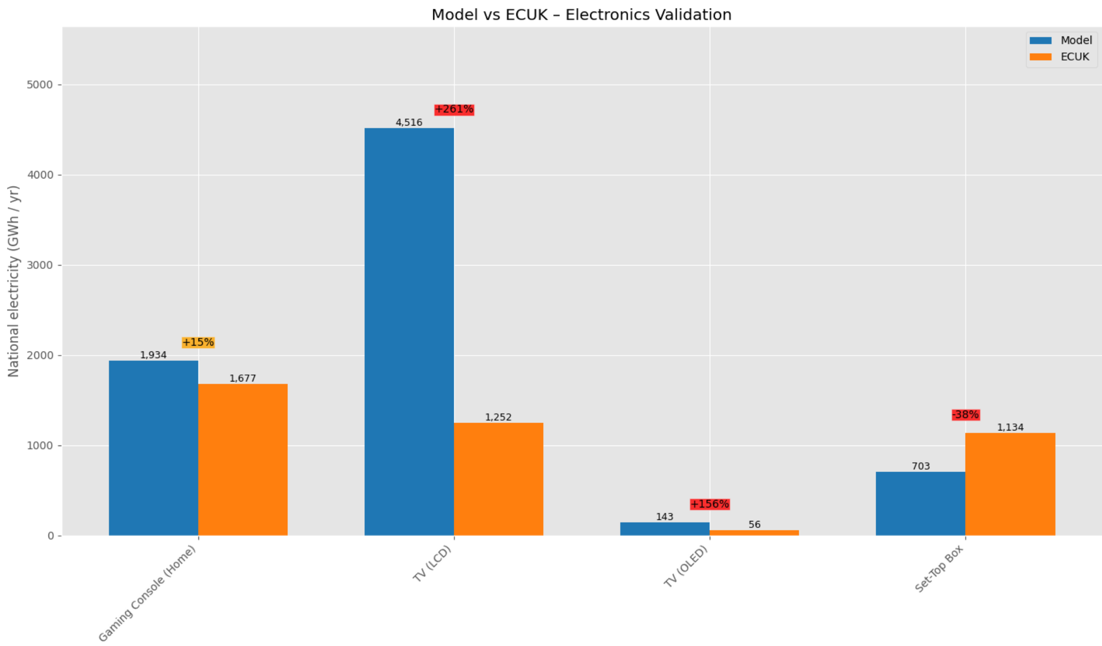

FIGURE 5.21: MODEL VALIDATION: COMPARISON OF NATIONAL ELECTRICITY CONSUMPTION FOR ENTERTAINMENT AND MEDIA DEVICES

The validation shows a reasonable alignment for Gaming Consoles (+15.3%), but extreme variances for other key devices, which can be analysed by considering the assumptions made in this model:

- **TV (LCD) (+261%):** The model estimates 4.5 TWh/yr, whereas the ECUK benchmark is 1.25 TWh/yr. This significant divergence occurs despite both analyses utilising a comparable stock base of approximately 52 million LCD sets derived from ECUK data. The primary point of divergence lies in the usage assumptions; this model assumes a daily viewing time of 4.5 hours per device, based on current data for 2025 from the UK regulator, Ofcom [Ofcom, 2025]. The discrepancy may be attributed to two potential factors. Firstly, while the power calculation in this model is systematic, it may not fully capture the wide variance in power ratings across all commercially available television models of the same size. Secondly, the ECUK benchmark may be founded on older, less representative viewing-time data, whereas this model employs the most up-to-date statistics available, which could explain the under-prediction in the benchmark figure.

- **TV(OLED) (+156%):** While the percentage difference is large, the low ownership of OLED TVs (approx. 1 million units) means the absolute difference in national energy is minor (a variance of 87 GWh/yr). From a national policy perspective, this discrepancy is less significant than that of the far more common LCD models.

- **Set-Top Box (-38.0%):** The under-prediction could be due to this model's assumption of a single 'on' power state (20.1W). This does not account for different operational modes, such as peak usage during recording versus idle-on time. The ECUK figure may be based on a profile that includes longer periods in higher-power active states or reflects an older, more diverse, and less efficient stock of devices still in use across the country, compared to the model’s efficient Sky Q set-up box, as the input parameters came purely from here.

5.5 Monte Carlo Analysis

To move beyond a purely deterministic result and understand the robustness of the model's conclusions, a Monte Carlo simulation was performed. This analysis is important as it quantifies how the inherent variability in real-world appliance power ratings affects the certainty of the final national energy estimates. The methodology, outlined in section 4.2, propagates this uncertainty through 10,000 runs to generate a probability distribution for the total national energy demand for each category. The results are presented in two ways: a histogram showing the probability distribution of the total national energy consumption, from which a median and a 90% confidence interval (P5 to P95) can be derived, which represents the range where the true value has a 90% probability of lying; and an error-bar plot showing the 90% confidence interval for the annual energy consumption of each household appliance.

5.5.1 Kitchen and Cooking Appliances

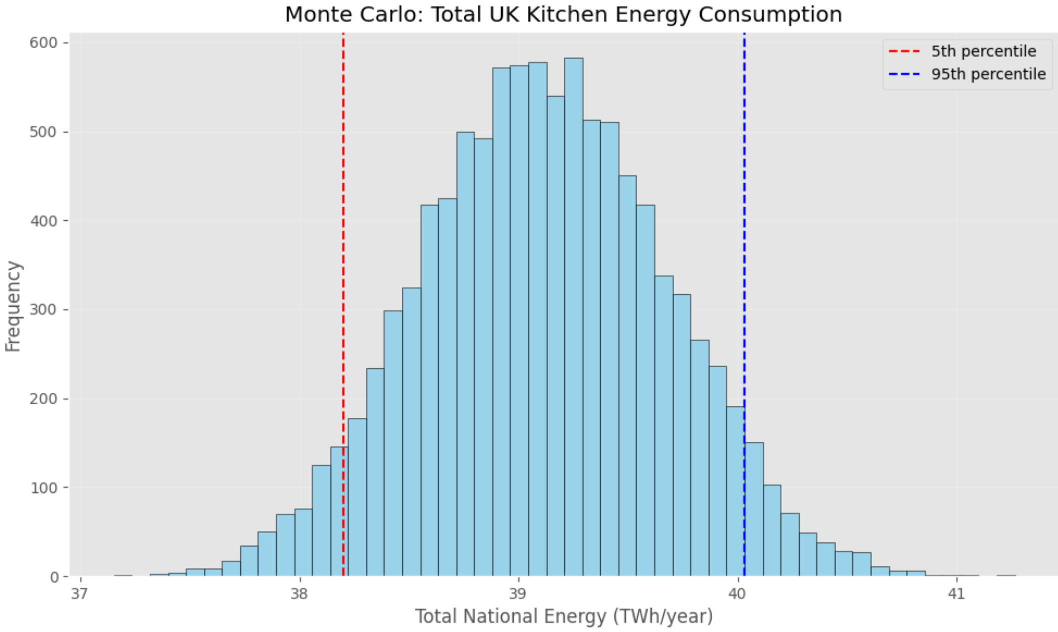

FIGURE 5.22: MONTE CARLO SIMULATION: PROBABILITY DISTRIBUTION FOR TOTAL UK KITCHEN ENERGY CONSUMPTION (TWH/YEAR)

The resulting distribution for the combined kitchen appliances is shown in Figure 5.22. The simulation indicates a median national consumption of approximately 39.2 TWh/year. The result lies within a 90% confidence interval (P5 to P95) spanning from 38.2 TWh/year to 40.0 TWh/year. This range of 1.8 TWh (or approx. ±2.3% of the median) quantifies the uncertainty in the final estimate, driven by the variability in the power ratings of millions of individual appliances.

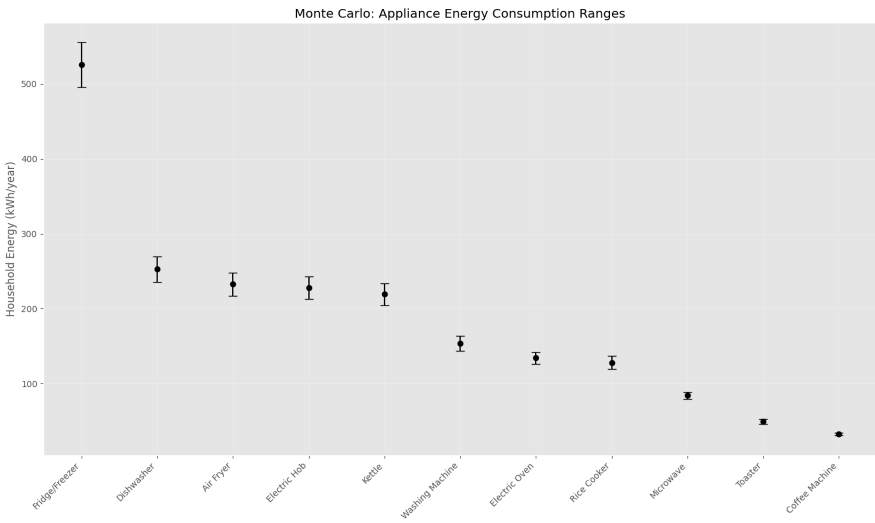

FIGURE 5.23: MONTE CARLO SIMULATION: 90% CONFIDENCE INTERVAL FOR PER-HOUSEHOLD KITCHEN APPLIANCE ENERGY CONSUMPTION (KWH/YEAR)

This national uncertainty is derived from the uncertainty at the individual appliance level, as shown in Figure 5.23. The error bars represent the 90% confidence interval for the annual energy consumption of a single household appliance. The Fridge/Freezer exhibits the widest uncertainty range, not because of a wider percentage of uncertainty, but because the ±10% variation is applied to a much larger base consumption value (525.6 kWh/yr), resulting in a larger absolute uncertainty. In contrast, devices with lower annual consumption, such as the Toaster and Coffee Machine, exhibit tighter clusters, as the ±10% variation results in a smaller absolute range.

5.5.2 Computing and Networking Devices

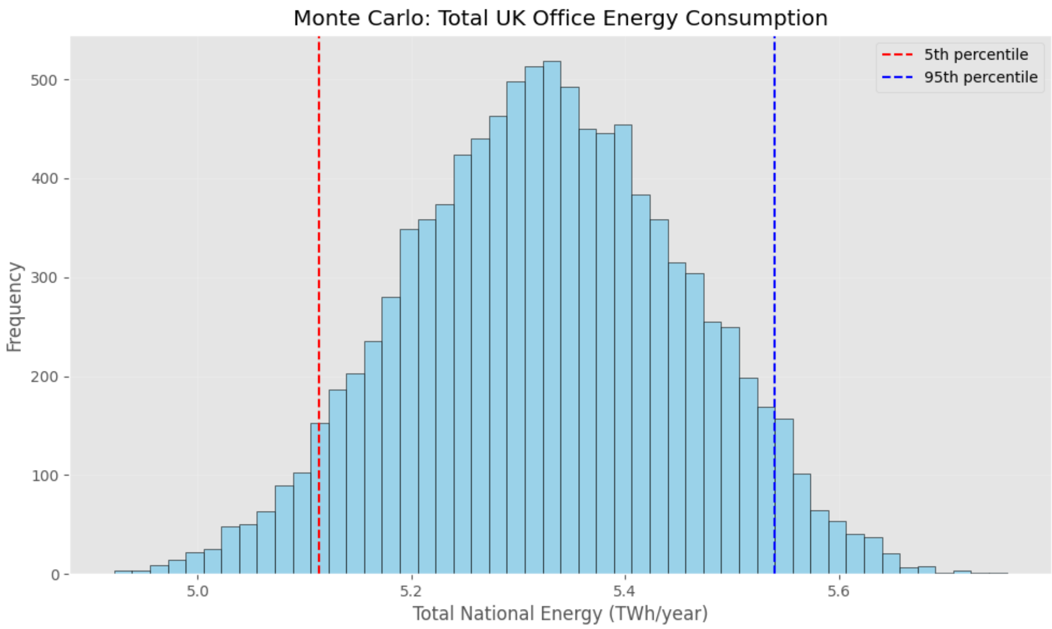

FIGURE 5.24: MONTE CARLO SIMULATION: PROBABILITY DISTRIBUTION FOR TOTAL UK OFFICE ENERGY CONSUMPTION (TWH/YEAR)

The resulting distribution for the combined computing and networking devices is shown in Figure 5.24. The simulation indicates a median national consumption of approximately 5.3 TWh/year. The result lies within a 90% confidence interval (P5 to P95) spanning from 5.1 TWh/year to 5.5 TWh/year. This range of 0.4 TWh (or approx. ±3.8% of the median) is primarily influenced by devices with both high consumption and a wide range of available models, such as routers and desktop computers.

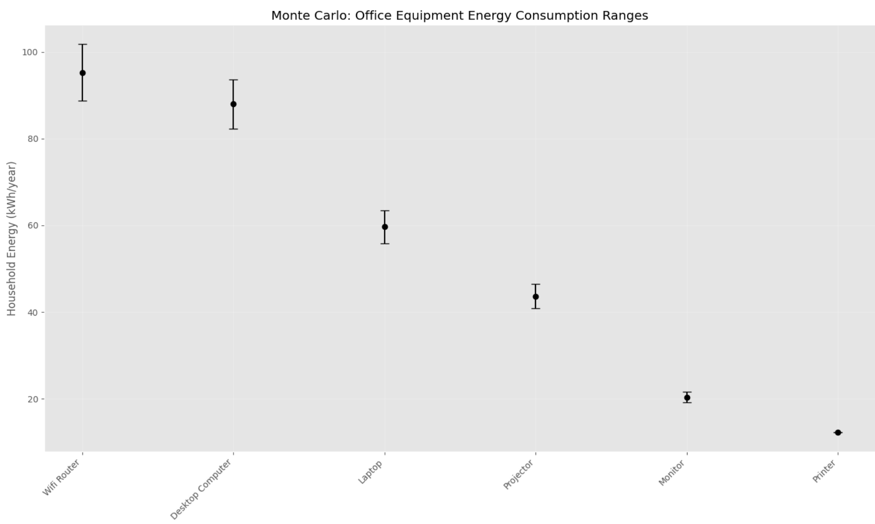

FIGURE 5.25: MONTE CARLO SIMULATION: 90% CONFIDENCE INTERVAL FOR PER-HOUSEHOLD OFFICE EQUIPMENT ENERGY CONSUMPTION (KWH/YEAR)

This national uncertainty is derived from the uncertainty at the individual appliance level, as shown in Figure 5.25. The Wifi Router and Desktop Computer exhibit the widest absolute uncertainty ranges, as the ±10% variation is applied to a larger base consumption value.

5.5.3 Personal Use Devices

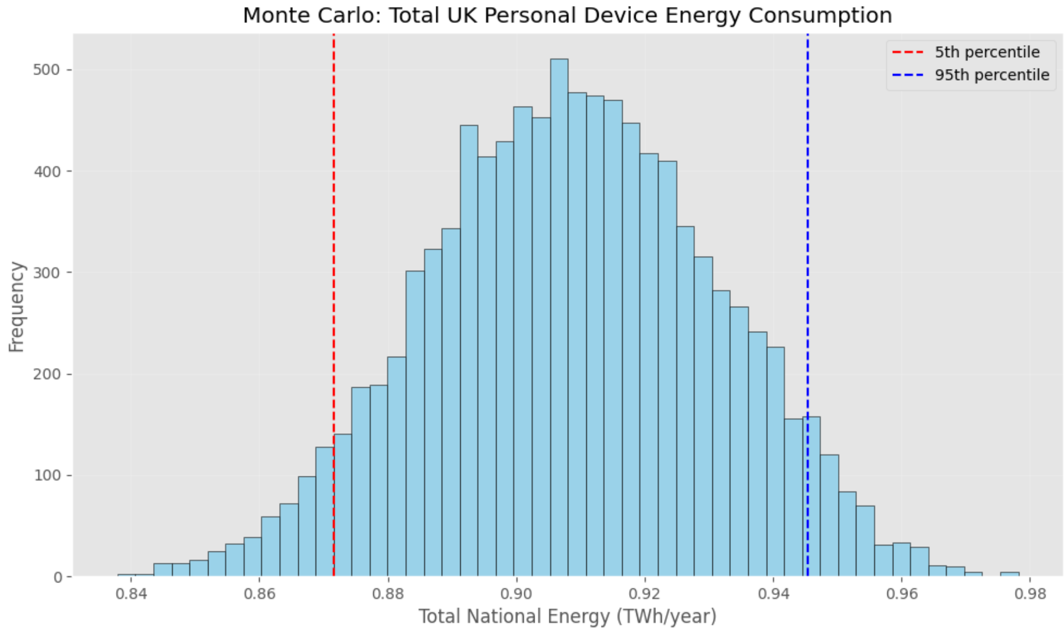

FIGURE 5.26: MONTE CARLO SIMULATION: PROBABILITY DISTRIBUTION FOR TOTAL UK PERSONAL DEVICE ENERGY CONSUMPTION (TWH/YEAR)

The resulting distribution for personal use devices is shown in Figure 5.26. The simulation indicates a median national consumption of approximately 0.91 TWh/year. The result lies within a 90% confidence interval (P5 to P95) spanning from 0.87 TWh/year to 0.95 TWh/year. This range of 0.08 TWh (or approx. ±4.4% of the median) is influenced by the high ownership and power variability of tablets and smartphones.

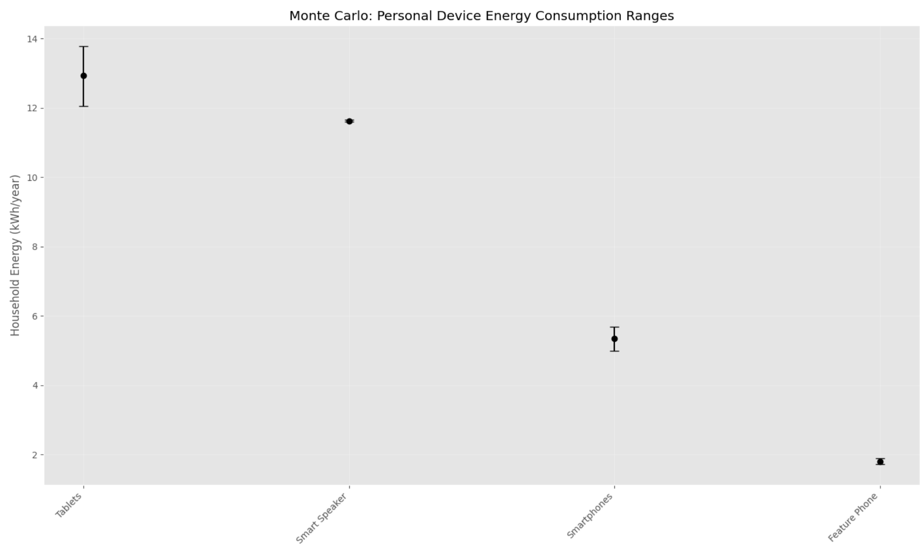

FIGURE 5.27: MONTE CARLO SIMULATION: 90% CONFIDENCE INTERVAL FOR PER-HOUSEHOLD PERSONAL DEVICE ENERGY CONSUMPTION (KWH/YEAR)

The national uncertainty is derived from the per-household uncertainty shown in Figure 5.27. Tablets exhibit the widest absolute uncertainty range, as the ±10% variation is applied to the highest base consumption in this category (12.9 kWh/yr).

5.5.4 Entertainment and Media Devices

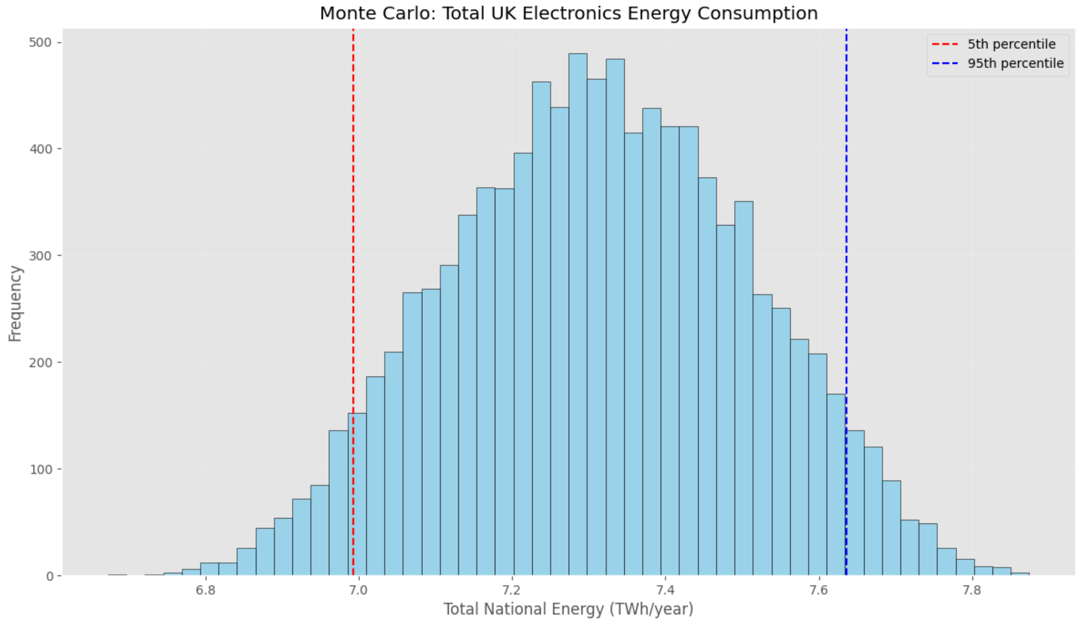

FIGURE 5.28: MONTE CARLO SIMULATION: PROBABILITY DISTRIBUTION FOR TOTAL UK ELECTRONICS ENERGY CONSUMPTION (TWH/YEAR)

Figure 5.28 summarises the Monte-Carlo uncertainty analysis for entertainment and media devices. Across 10,000 random draws, the distribution centres on ~ 7.3 TWh/yr; the 5th and 95th percentiles fall at ~ 7.0 TWh and ~ 7.6 TWh, giving a 90 % confidence range of about 0.6 TWh.

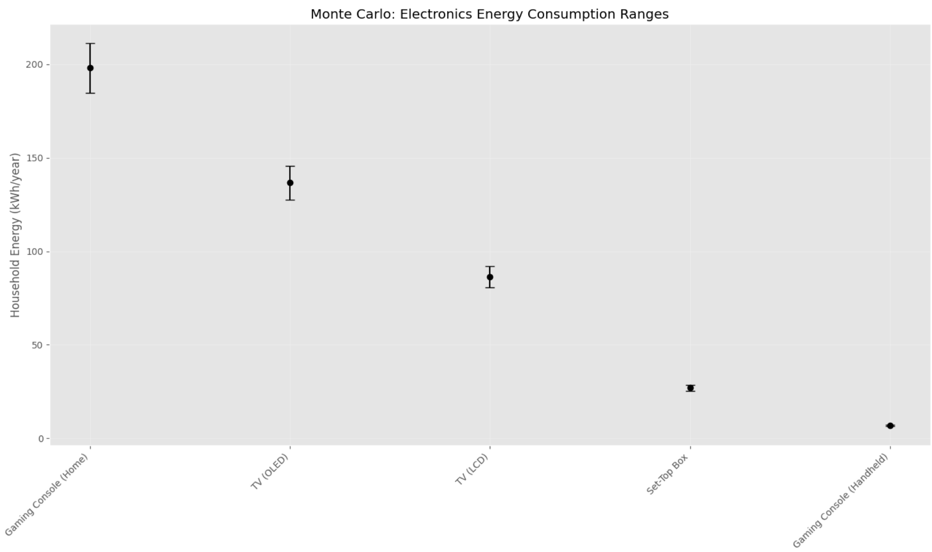

FIGURE 5.29: MONTE CARLO SIMULATION: 90% CONFIDENCE INTERVAL FOR PER-HOUSEHOLD ELECTRONICS ENERGY CONSUMPTION (KWH/YEAR)

The national uncertainty is derived from the per-household uncertainty shown in Figure 5.29. The Home Gaming Console exhibits the widest absolute uncertainty range because the ±10% variation is applied to the highest base consumption in this category (198.0 kWh/yr).

5.6 Aggregate Results and Sensitivity Analysis

This final section synthesises the results from all device categories to provide a holistic overview of the model's findings. It examines the aggregate national consumption and identifies the primary drivers of energy use at both the category and device-level. It concludes with a sensitivity analysis to determine which input parameters have the most significant influence on the model's overall outputs. This is crucial for understanding the model's robustness and for guiding policymakers towards the areas where improved data collection or targeted interventions would have the most impact.

5.6.1 Overall Consumption by Category

Summing all 26 devices, the model estimates a total annual operational energy consumption of 52,659.6 GWh, resulting in 11,866.8 kt of CO_2e.

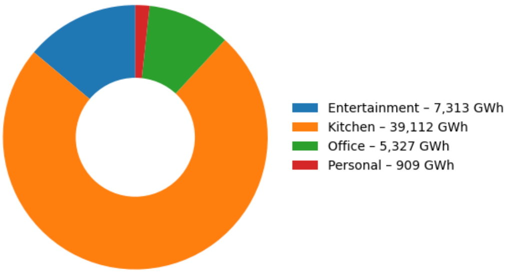

FIGURE 5.30: AGGREGATE ANNUAL ENERGY CONSUMPTION BY APPLIANCE CATEGORY (GWH)

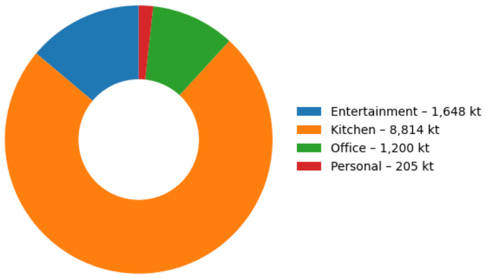

FIGURE 5.31: AGGREGATE ANNUAL CO_2E EMISSIONS BY APPLIANCE CATEGORY (KT)

The Kitchen and Cooking category remains the dominant load, now accounting for ~ 39.1 TWh/yr (~74.3% of the total). Its lead is still driven by a mix of high-power heating devices and continuously running refrigeration. Entertainment and Media follow with ~ 7.3 TWh/yr (~13.9 %), largely reflecting the ubiquity and long usage hours of televisions. The Office (computing and networking) category contributes ~5.3 TWh/yr (~10.1 %), while Personal-Use devices add just ~ 0.9 TWh/yr (~1.7 %). The emissions split in Figure 5.31 mirrors these percentages almost exactly, reinforcing that Kitchen and Entertainment appliances remain the primary targets for national efficiency policy.

5.6.2 Key Appliance Contributions within Categories

While the category-level analysis is useful, a deeper insight is gained by examining which specific devices dominate within each group. The doughnut charts in Figure 5.26 provide this granular breakdown.

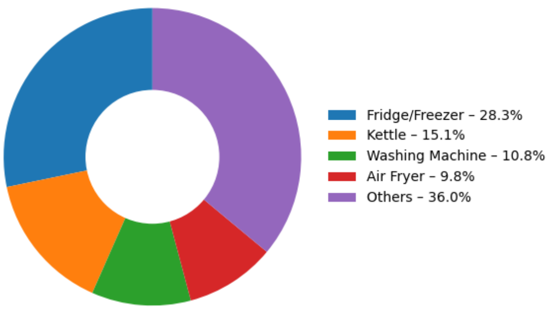

FIGURE 5.32: CONTRIBUTION OF KEY APPLIANCES TO TOTAL KITCHEN ENERGY CONSUMPTION (%)

Kitchen devices

They remain heavily dominated by refrigeration and water-heating loads. Fridge-freezers consume about 11 TWh/yr, while kettles add a further 5.9 TWh/yr; together they represent 43 % of all kitchen electricity in the model. Adding washing machines at 4.2 TWh/yr lifts the share of just three appliances to 54 %, and the rapidly popular air-fryer already contributes almost 3.9 TWh/yr (10 %). Because these high-duty devices account for nearly two-thirds of kitchen demand, tightening efficiency standards for refrigeration, improving kettle and washing-machine labelling, and promoting behaviour change around these uses will deliver the quickest savings. The eleven remaining appliances, shown collectively as “Others” at roughly 14 TWh/yr, are a secondary target once the headline loads have been addressed.

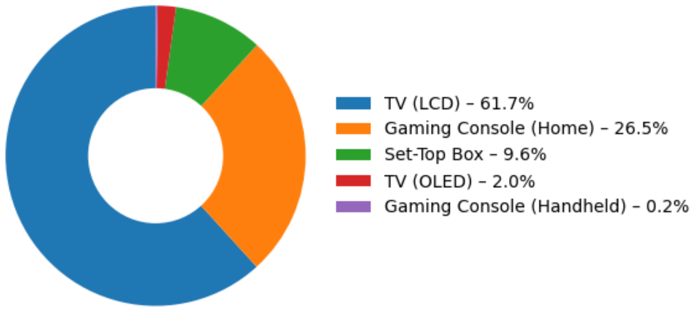

FIGURE 5.33: CONTRIBUTION OF KEY APPLIANCES TO TOTAL ENTERTAINMENT ENERGY CONSUMPTION (%)

Entertainment devices

They remain highly concentrated in one appliance group. LCD televisions account for about two-thirds of the category’s electricity (~ 61.7 %), underlining how screen size and long viewing hours shape national demand. Home gaming consoles follow at ~ 26.5 %, while set-top boxes add a further ~ 9.6 %. OLED TVs and handheld consoles together contribute less than 2.5 %, reflecting their still-limited market share. The large share of energy consumption attributed to televisions in this model is subject to considerable uncertainty. This uncertainty arises from two conflicting possibilities. On one hand, the model may present an over-prediction. This is a potential consequence of the methodology, which applies a single weighted-average on-mode power value to all televisions within a specific size category and may not fully account for the market prevalence of more energy-efficient models. On the other hand, the discrepancy could stem from an under-prediction by the ECUK benchmark, which may be based on less current data. Clarifying this is critical for policy direction; for instance, if a more granular analysis of market data revealed that the prevalence of highly efficient models results in a lower fleet-average power rating than assumed, the television's overall energy share would be smaller, and policy attention would necessarily shift toward devices like gaming consoles, whose power consumption is expected to rise with successive hardware generations.

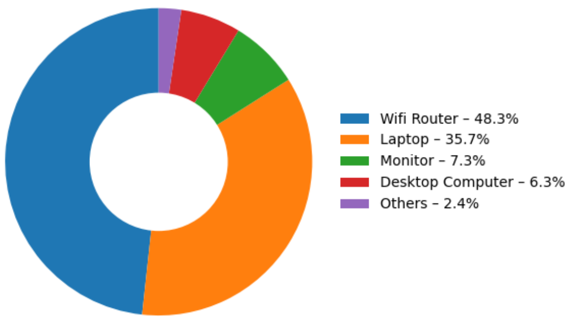

FIGURE 5.34: CONTRIBUTION OF KEY APPLIANCES TO TOTAL OFFICE ENERGY CONSUMPTION (%)

Office (computing) devices

Demand is still driven by two practically "always-on" devices. Wi-Fi routers account for 48 % of the category's electricity, and laptops add another 36 %, so together they make up just over four-fifths of office consumption. The laptop share reflects both its large battery (~ 52 Wh) and the 3 h 39 m average daily charging window assumed in the model. Combined, these two devices come to roughly 4.5 TWh/yr and one million tonnes of CO_2e, underscoring how continuous connectivity and frequent charging dominate energy use in the home-office set-up.

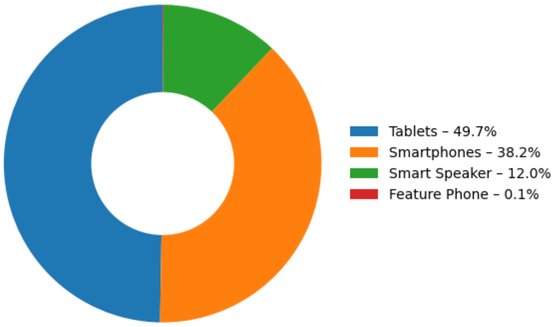

FIGURE 5.35: CONTRIBUTION OF KEY APPLIANCES TO TOTAL PERSONAL DEVICE ENERGY CONSUMPTION (%)

Personal devices

The demand is shaped almost entirely by the ubiquity of two rechargeable products. Tablets make up 49.7 % of the category and smartphones 38.2 %, so together they account for around 88 % of personal-electronics electricity use. Their dominance is not due to high per-unit power; each charges from a modest 5-18 watt adapter, but to scale: roughly one hundred million tablets and phones are plugged into UK sockets every day. A third device, the smart speaker, adds 12 % through near-constant stand-by listening, while feature phones now contribute only 0.1 %. The result underlines that, for small gadgets, ownership volume rather than individual wattage drives national load, pointing policy toward mass-market charging standards and low-loss adapters rather than bespoke efficiency measures for niche devices.

5.6.3 Sensitivity Analysis

A one-at-a-time (OAT) local sensitivity analysis was carried out on all 78 input parameters (power, usage time and ownership for 26 devices). Each parameter was perturbed by ±10 % while the others were held constant, and the resulting change in the national total was recorded.

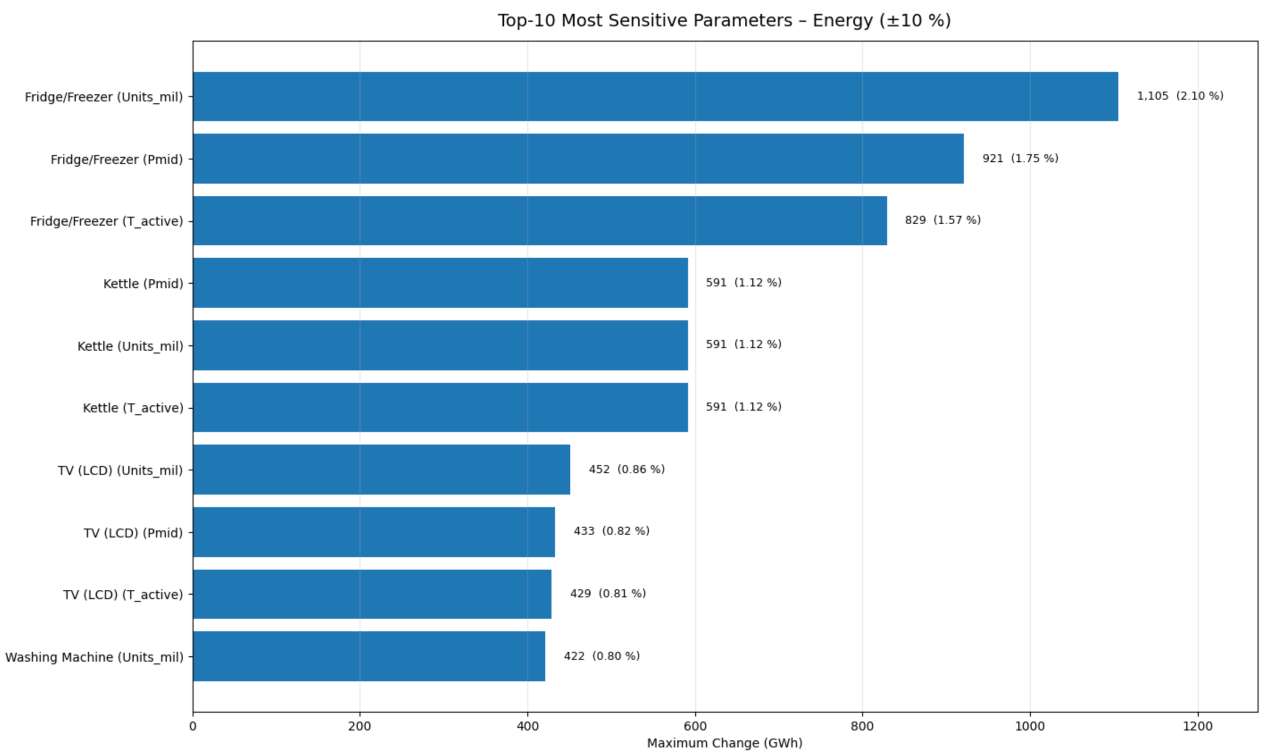

FIGURE 5.36: SENSITIVITY ANALYSIS: TOP TEN MOST SENSITIVE INPUT PARAMETERS ON NATIONAL ENERGY CONSUMPTION

The model is most sensitive to variables linked to the two highest-consuming appliance types. A ±10 % change in Fridge/Freezer ownership (Units_mil) shifts the national UK energy consumption by 1,105 GWh, about 2.06 % of the 52,659.6 GWh baseline. The next-ranked drivers are the power rating (Pmid) and active time (T_active) of Fridge/Freezers, followed by the power rating, ownership, and active time of Kettles. Altogether, the six parameters for these two devices outweigh the influence of the remaining 72 inputs combined, so model accuracy rests primarily on reliable refrigeration and kettle data.

Kettles appear in the top ten because their consumption is the product of three large numbers: 3 kW power × 27 million units × 12 min per day. A change in any one of those inputs is amplified by the other two, making the device highly influential despite short runtimes.

LCD-TV dominance is also notable, with changes to ownership (Units_mil), power (Pmid), and screen-on time (T_active) all appearing in the top ten. However, the primary source of uncertainty in these calculated shares is believed to be the power parameter, not the usage data. This uncertainty stems from two possibilities: either this model over-predicts consumption by using a single weighted-average power rating that may not fully represent the prevalence of more energy-efficient models, or the ECUK benchmark under-predicts the figure by relying on potentially outdated data. Verifying real-world usage with smart-meter or BARB-style viewing data, and instrumenting fridge compressor duty cycles, would therefore deliver the greatest reduction in model uncertainty.

The CO_2e sensitivity plot mirrors the energy results: the 0.225 kg/ kWh conversion means a ±1,105 GWh swing translates directly into a ±249 kt CO_2e shift.

5.7 Fulfilment of Objectives and Success Criteria

This chapter has detailed the quantitative outputs of the energy simulation model and, in doing so, has successfully fulfilled the three core objectives laid out in the problem formulation. The first objective, to compile a validated stock database for 26 devices, was met through the detailed parameter selection in the methodology. The second, the implementation of a Python-based model to output energy and emissions data, has been demonstrated through the extensive national and household-level results presented throughout this chapter; the core calculations used in the model can be seen in section 4.2 and appendices A and B. Finally, the third objective was achieved by ranking appliances to identify high-priority devices for policy intervention. The analysis confirmed that a 20% reduction in consumption for the top national emitters would yield significant carbon savings, specifically for the Fridge/Freezer (~0.50 Mt CO_2e/yr), Kettle (~0.27 Mt CO_2e/yr), and TV (LCD) (~0.203 Mt CO_2e/yr), all of which comfortably exceed the 0.2 Mt CO_2e/yr target and thus stand out as critical targets for evidence-based efficiency regulations.

Furthermore, the project has satisfied all three predefined success criteria, validating the robustness of these findings. Internal coherence was established by successfully reconciling all device-level stock, power, and usage inputs into a comprehensive national total. The criterion for external plausibility was met through the rigorous validation against the ECUK dataset, where significant variances were explained by clearly documented differences in methodological approaches and underlying behavioural or technical assumptions. Lastly, uncertainty transparency was achieved by propagating input variability through a 10,000-draw Monte Carlo analysis and reporting the resulting P5-P95 confidence bounds alongside each headline result, ensuring the model's certainty was clearly and robustly communicated. Together, the fulfilment of these objectives and criteria validates the model as a potent tool for granular, bottom-up energy analysis, setting the stage for the policy implications discussed in the final chapter.

Chapter 6: Conclusion

6.1 Synthesis of the Research: Addressing the Granularity Gap

This project addressed a critical "granularity gap" in the UK's energy evidence base, where official statistics lack the device-specific detail needed for effective appliance regulation. To fill this gap, a transparent, bottom-up engineering model was developed to quantify the electricity consumption and emissions of 26 common household appliances, offering a more robust and adaptable methodological alternative to existing top-down frameworks. This approach provides the granular insights necessary to formulate targeted policies and support the UK's net-zero ambitions.

6.2 Principal Findings and Achievement of Research Objectives

The application of the bottom-up model yielded a detailed, multi-faceted view of UK household energy consumption. The principal findings revealed a total annual consumption of approximately 53.7 TWh for the 26 devices, with a clear hierarchy dominated by Fridge/Freezers (11.1 TWh), Kettles (5.9 TWh), and LCD Televisions (4.5 TWh). The sensitivity analysis further confirmed that the model's aggregate national total is most sensitive to the input parameters for these top-ranked appliances, highlighting where data accuracy is most critical.

These findings successfully answered the project's core research questions by quantifying appliance energy use, demonstrating how a deterministic model can derive these values, showing the sensitivity of the rankings, and confirming why appliance-level insights are crucial for targeted regulation. In doing so, the study also fulfilled all three of its primary objectives. A validated stock database was compiled, the Python-based energy model was successfully implemented, and a ranking of appliances by emissions confirmed that a 20% efficiency improvement in Fridge/Freezers, Kettles, and LCD Televisions would yield annual carbon savings that comfortably exceed the 0.2 Mt CO_2e target.

6.3 Implications of the Findings for Policy, Industry, and Consumers

The findings of this study carry significant implications. The research provides a data-driven hierarchy of impact, revealing non-intuitive results such as Kettles contributing more to the national load than Washing Machines. This enables policymakers to target high-consumption devices like Fridge/Freezers with national standards, while consumer-facing campaigns can focus on behaviour related to high-intensity appliances like Dishwashers. The model also serves as a critical check on official benchmarks; its purely bottom-up methodology provides an alternative perspective to the hybrid ECUK framework, suggesting that the impact of key appliance categories may be underestimated by current statistics. Furthermore, the study highlights the growing importance of the "always-on" digital home, with devices like Wi-Fi routers contributing significantly to baseline demand, pointing to a need for future policies to address networked standby power.

6.4 Critical Reflection on the Study's Limitations

In the interest of academic rigour, it is essential to acknowledge the study's limitations. The model is built upon deterministic behavioural assumptions, using fixed average usage times that do not capture real-world variability. The scope is focused on operational emissions, excluding the embodied carbon from manufacturing and disposal, and does not yet incorporate macroeconomic feedbacks like price elasticity. Finally, the validation process was itself constrained by the nature of the ECUK benchmark; reconciling this study's granular results with the official national statistics was challenging due to the latter's aggregated product categories and reliance on older behavioural data.

6.5 Recommendations for Future Research

These limitations inform a clear agenda for future research. The highest priority is to enhance model accuracy by gathering fresh, empirical UK run-time data through smart-plug trials or updated Time-Use Surveys. Future work should also focus on integrating an economic module to capture price elasticity and rebound effects, and expanding the model's scope to include more granular operational states and embodied emissions for a full lifecycle assessment. The sensitivity analysis provides a guide for this work, indicating that data collection should be prioritised for fridge-freezers, kettles and televisions. This is particularly critical for LCD televisions, where the validation against ECUK data revealed a significant (+261%) discrepancy. Further research is required to resolve this variance and ascertain whether it stems from an over-prediction by this model, driven by its power rating assumptions, or a significant under-prediction by the ECUK benchmark.

6.6 Concluding Remarks: A Foundational Tool for UK Energy Analysis

This project has developed and validated a transparent, bottom-up model of UK household appliance energy consumption, filling a critical evidence gap. The research provides an immediately useful tool for policymakers and a robust foundation for the next generation of analysis required to support the UK's transition to a net-zero future.

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Appendix A

This calculation determines a single, representative power figure for a household with a primary and secondary LCD television, each with different power ratings and daily usage times.

1. Input Data

Primary sets (main TV)

- Stock = 26.3 million
- Screen-size mix → on-mode power

  17% ≤ 32″ → 41 W
  23% ~ 37″ → 50 W
  27% 43–50″ → 72 W
  20% ≥ 55″ → 80 W

- Weighted on-mode power

  (0.17×41 + 0.23×50 + 0.27×72 + 0.20×80) / 0.87 = 63.7 W

Secondary sets (second TV)

- Stock = 26.1 million
- Screen-size mix → on-mode power

  20 % ≤ 40″ → 41 W
  66 % < 50″ → 72 W
  14 % ≥ 55″ → 80 W

- Weighted on-mode power

  (0.20×41 + 0.66×72 + 0.14×80) / 1.00 = 36.2 W

Stand-by power (both groups) = 0.5 W

Total TV usage time = 270 min per day

Fleet-average on-mode power (stock-weighting)

$$P_{\text{fleet}} = \frac{N_p P_p + N_s P_s}{N_p + N_s} = \frac{26.3\,\text{M} \times 63.7\,\text{W} + 26.1\,\text{M} \times 36.2\,\text{W}}{52.3\,\text{M}} = \mathbf{50.4\ W}$$

This 50.4 W value feeds directly into the national energy calculation. Stand-by energy is added separately using 0.5 W across 24 h.

Appendix B

Battery-device energy calculation

- C_{bat}= battery capacity (Wh)
- P_{mid} = charger power rating (W)
- P_{stby} = idle-charger draw (W)
- n_{cycles} = full-charge cycles per day (assumed 1)

<u>Daily active charge time (minutes per day)</u>

$$T_{\text{act}} = \frac{C_{\text{bat}} \times n_{\text{cycles}}}{P_{\text{mid}}} \times 60$$

<u>Annual active-use energy (kWh / device / yr)</u>

$$E_{act,device} = C_{bat} \times n_{cycles}$$

Annual idle-draw energy (kWh / device / yr)

$$E_{\text{stby,device}} = \frac{P_{\text{stby}}}{1000} \times \frac{1440 - T_{\text{act}}}{60} \times 365$$

<u>Total annual device-level energy</u>

$$E_{\text{device}} = E_{\text{act,device}} E_{\text{stby,device}}$$

Assumes zero charging losses and one full cycle per day; all other steps then follow the same equations used for mains-powered devices.

Appendix C

The Python code used for the model and analysis presented in this dissertation can be found in the following GitHub repository:

https://github.com/Yxzid/Yazid-Askar-Dissertation-Codes.git

This repository includes all scripts, data processing workflows, and any other relevant code used to generate the results and figures in this work, ensuring reproducibility and transparency.